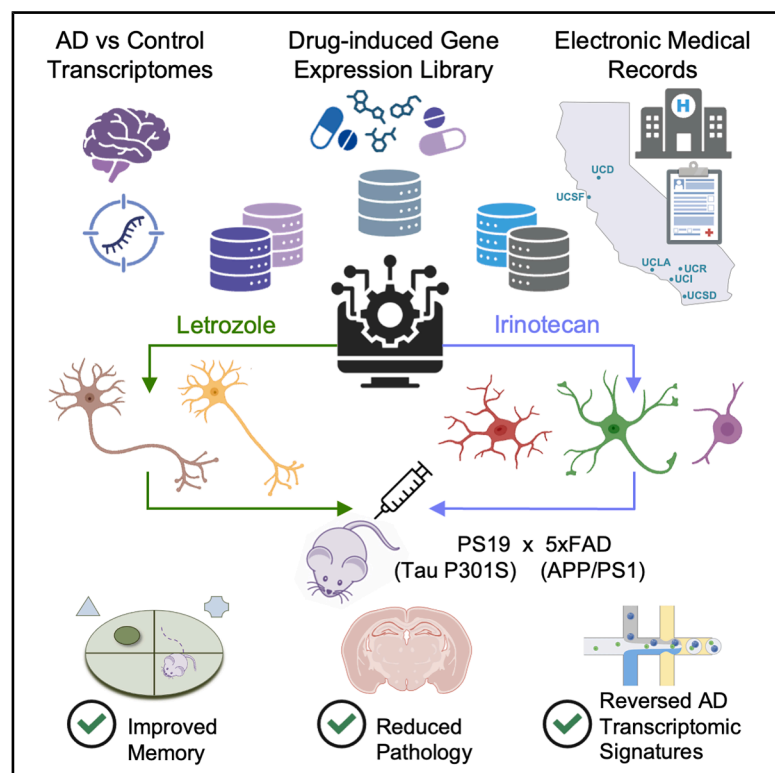


Cell-type-directed network-correcting combination therapy for Alzheimer's disease

Graphical abstract



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In brief

A multi-cell-type drug discovery strategy targeting dysregulated gene networks in neurons and glia identified letrozole and irinotecan as a combination therapy that significantly improved memory and reduced pathology in preclinical Alzheimer's models.

Highlights

- Human transcriptomic and drug repurposing analyses identify letrozole and irinotecan for AD
- Real-world EMR data reveal lower AD risks in patients exposed to letrozole or irinotecan
- Letrozole and irinotecan combination therapy improves memory and pathology in AD models
- Transcriptomic reprogramming represents a promising strategy for treating complex diseases



Article

Cell-type-directed network-correcting combination therapy for Alzheimer's disease

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SUMMARY

Alzheimer's disease (AD) is a multifactorial neurodegenerative disorder characterized by heterogeneous molecular changes across diverse cell types, posing significant challenges for treatment development. To address this, we introduced a cell-type-specific, multi-target drug discovery strategy grounded in human data and real-world evidence. This approach integrates single-cell transcriptomics, drug perturbation databases, and clinical records. Using this framework, letrozole and irinotecan were identified as a potential combination therapy, each targeting AD-related gene expression changes in neurons and glial cells, respectively. In an AD mouse model with both A β and tau deposits, this combination therapy significantly improved memory performance and reduced AD-related pathologies compared with vehicle and single-drug treatments. Single-nucleus transcriptomic analysis confirmed that the therapy reversed disease-associated gene networks in a cell-type-specific manner. These results highlight the promise of cell-type-directed combination therapies in addressing multifactorial diseases like AD and lay the groundwork for precision medicine tailored to patient-specific transcriptomic and clinical profiles.

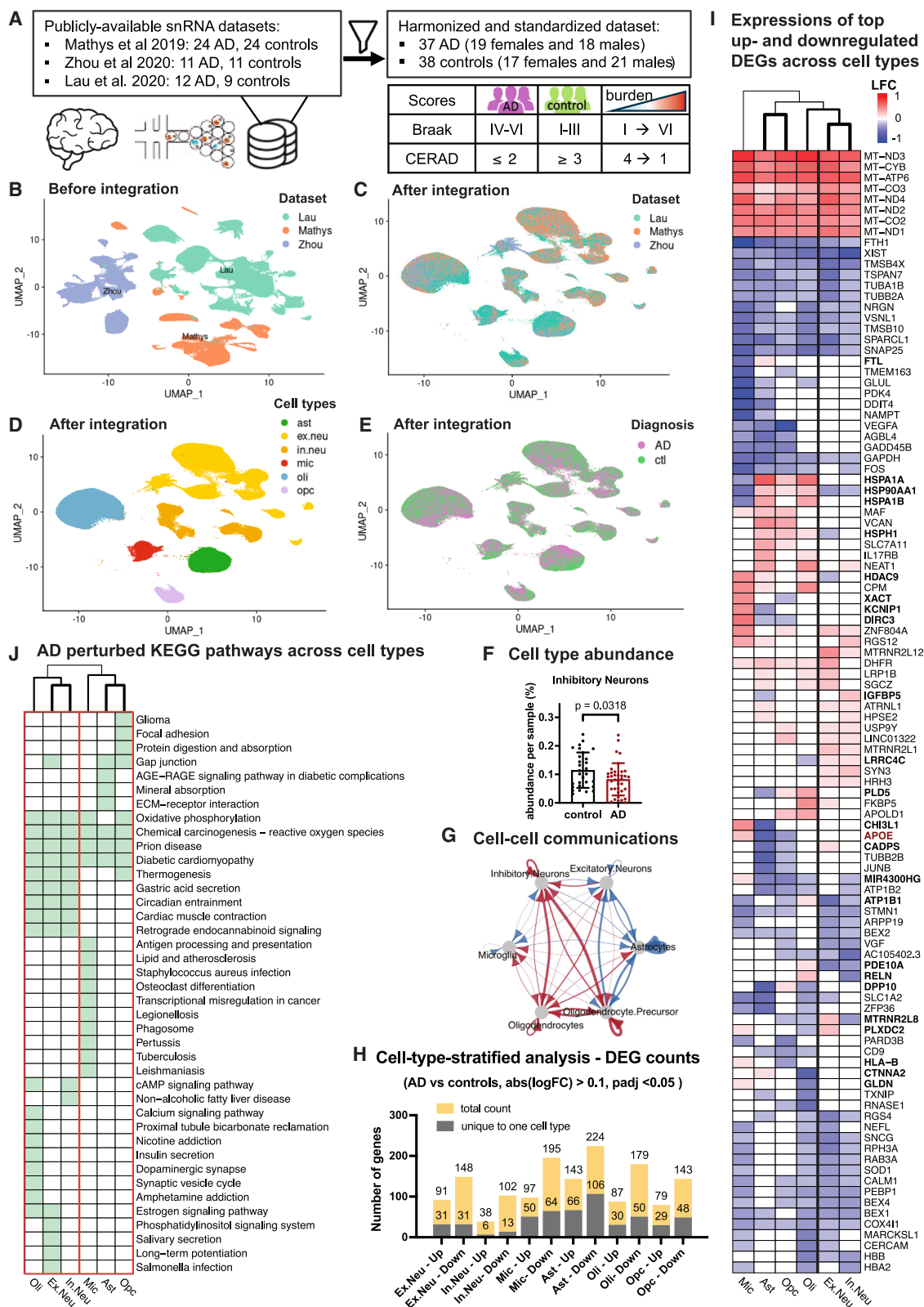
INTRODUCTION

Alzheimer's disease (AD) is a neurodegenerative disorder with severe impacts on individuals, families, and society,¹ and yet without a cure. AD patients suffer from progressive memory loss, behavior changes, and cognitive and motor deficits, leading to diminished quality of life.² Over 50 million people are currently living with AD or other related dementias worldwide, a number projected to triple by 2050.^{3,4} With global costs exceeding \$1 trillion annually, AD is one of the costliest health conditions worldwide.⁵ Urgent action is needed to develop more effective and accessible treatments.

Despite rigorous preclinical and clinical research efforts, AD drug development faces significant challenges, with a 98% failure

rate in recent decades.⁶ Current therapeutic options are mostly limited to symptom-managing treatments.⁷ Furthermore, recently approved immunotherapies have only modest effects on disease progression.^{8,9} The lack of effective treatments stems from the pathological heterogeneity in AD.¹⁰ Although AD's most prominent disease hallmark is proteopathy, characterized by extracellular amyloid- β (A β) plaques and intracellular tau neurofibrillary tangles (NFTs), their interplay and exact mechanisms leading to disease remain unclear.^{11,12} Genetic heterogeneity further complicates the disease, including risk mutations in the amyloid precursor protein (APP) and presenilin (PSEN) genes, and APOE4, a risk isoform of apolipoprotein E (APOE).¹³ More recently, emerging evidence suggests a shift in focus toward a more complex cellular phase of AD.¹⁴ Traditionally, AD research has been dominated





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by a neuron-centric model, emphasizing molecular mechanisms that drive synaptic dysfunction, neuronal loss, and neurodegeneration.^{11,15} However, recent studies highlight the critical role of glial cell dysfunction in disease progression, with processes such as neuroinflammation, metabolic impairment, and demyelination driven by malfunctioning microglia, astrocytes, and oligodendrocytes.^{14,16,17} These suggest that both neuronal vulnerability and inadequate support from dysfunctional glial cells are integral to the heterogeneous mechanisms underlying AD.¹⁸ Considering the multifactorial nature of AD, traditional therapeutic approaches focusing on single disease hallmarks or bulk-tissue-level pathologies are often insufficient, leading to variable treatment outcomes.

Given the unmet need for disease-modifying treatments, drug repurposing has gained interest due to its faster development, lower costs, and improved safety.¹⁹ In addition, technical advancements in mining large-scale databases offer new opportunities for discovering promising candidates. These developments, combined with various drug-screening approaches, such as *in vitro*²⁰ and *in vivo* assays,²¹ gene signature matching,²² network modeling,²³ machine learning,²⁴ and data mining,²⁵ have identified numerous repurposed candidates for AD over the past decade. However, the extensive array of potential candidates complicates the establishment of priorities for clinical translation.²⁶

In this study, we proposed repurposing a combination of 2 existing drugs, letrozole²⁷ (an aromatase inhibitor used in the treatment of hormone-receptor-positive breast cancer) and irinotecan²⁸ (a topoisomerase inhibitor used to treat colorectal and pancreatic cancer), to reverse cell-type-specific gene expression alterations across multiple distinct cell types implicated in AD. Our drug-screening strategy is entirely driven by human data, utilizing large-scale omic datasets from post-mortem brains,^{29–31} a drug perturbation library generated in human cell lines,³² and clinical records encompassing millions of individuals, thereby maximizing the chance of clinical translation. We validated our predicted drug candidates through dosing experiments in an AD mouse model expressing both mutant human APP/PSEN1 and tau by crossing the 5×FAD³³ and PS19³⁴ lines, two widely used models that recapitulate many AD-related phenotypes.³⁵ We demonstrated that the combination treatment tar-

geting both neurons and glial cells significantly ameliorated memory deficits and AD-related pathologies compared with vehicle treatment and outperformed single-drug treatments targeting either neurons (letrozole) or glial cells (irinotecan) alone. The successful *in vivo* validation underscores the power of multi-cell-type network-correction therapies in effectively treating AD, highlighting the promise of our human-data-driven, cell-specific drug discovery approach to develop comprehensive therapies for complex diseases. Furthermore, by targeting molecular signatures and clinical features derived from real-world individuals, this study demonstrates the potential for AI-enabled precision medicine leveraging large-scale multimodal personalized measurements.

RESULTS

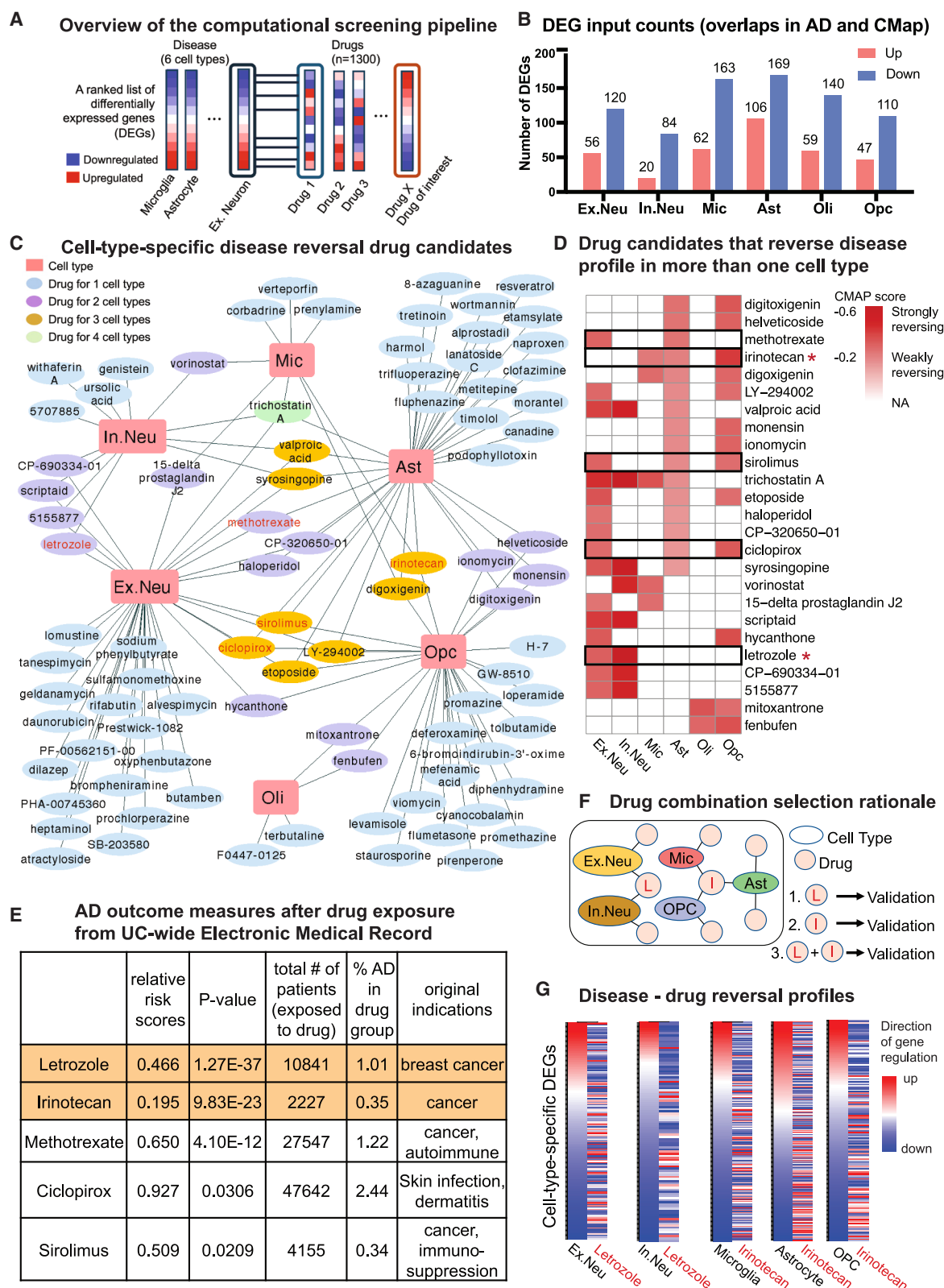
Computational integration of large-scale single-nucleus transcriptomic datasets from multiple studies enables the identification of cell-type-specific AD signature profiles

We organized a comprehensive single-nucleus RNA sequencing (snRNA-seq) dataset by combining published data from three independent studies^{29–31} (Figure 1A), thereby enhancing patient representation by incorporating samples from diverse biological and technical contexts across populations and regions.^{36,37} Case and control classification across studies was standardized using consortium to establish a registry for Alzheimer's disease (CERAD)³⁸ and Braak³⁹ scores, excluding samples not meeting both criteria (Figure 1A; STAR Methods). The uniform manifold approximation and projection (UMAP) visualization reveals clustering predominantly by the datasets of origin (Figures 1B and S1A). Effective computational integration identifies and corrects batch effects and technical artifacts that may arise during data collection in individual studies, ensuring data consistency and reliability.^{40,41} After batch harmonization with the integration algorithm, technical artifacts between datasets were effectively removed (Figure 1C) and distinct cell populations clustered by cell types (Figure 1D), although no discrete separation was observed based on disease status (Figure 1E).

To systematically characterize cell-type-specific AD pathophysiological features, we conducted comprehensive analyses

Figure 1. Single-nucleus transcriptomic profiling reveals both shared and cell-type-specific gene expression signatures in human AD samples

- (A) Data sources (publicly available) and cohort summary, including sample filtering criteria and case-control standardization strategies.
(B) UMAP plot of merged samples from three independent studies. Study-associated variations are apparent without integration.
(C–E) UMAP plots of harmonized dataset using Seurat canonical correlation analysis (CCA) integration algorithm, with study variation eliminated, and labeled by study (C), cell type identity (D), and case-control identification (E).
(F) Scattered bar plot of cell type abundance percentage for inhibitory neurons, comparing AD and controls (unpaired two-sided t test; data are represented as mean ± SEM).
(G) Projection plots for cell-type-to-cell-type communication (CCC) measured by expression of receptor and ligand pairs and represented as arrows connecting two cell types. Red connections indicate increased communication and blue indicate decreased communications comparing AD versus controls. Arrows point to the directions from sender to receiver of the communication.
(H) Bar plot of differentially expressed gene (DEG) counts of AD versus controls in the six major brain cell types and separated into up- and downregulated subgroups. Counts of DEGs unique to one cell type were superimposed on the total counts.
(I) Heatmap showing overlaps of top DEGs (highest absolute log fold changes) in each cell type; DEGs with opposite regulatory patterns in AD across cell types are bolded.
(J) Heatmap showing AD-enriched KEGG pathways across cell types. Few functional pathways are shared by all six cell types. Oligodendrocytes cluster with excitatory and inhibitory neurons, suggesting more similarity among these cell types, while microglia cluster with astrocytes and OPCs.
See also Figure S1 and Tables S1 and S2.



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focusing on six disease-relevant cell types: excitatory neurons (ex.neu), inhibitory neurons (in.neu), microglia (mic), astrocytes (ast), oligodendrocytes (oli), and oligodendrocyte precursor cells (OPCs). Among them, only the proportion of inhibitory neurons significantly decreased in AD patients compared with controls (Figures 1F and S1B), as previously reported in human and mouse models of AD.^{42–44} Cell-cell-communication (CCC) analysis revealed heterogeneous signaling patterns among neuronal subpopulations in AD compared with controls. Signaling to inhibitory neurons increased from all cell types except microglia, while signaling to excitatory neurons decreased from all cell types except oligodendrocytes (Figures 1G and S1C). These molecular insights suggest a disruption of the balance between excitatory and inhibitory neuronal activities in AD, potentially driven by altered neuronal signaling dynamics.

In our integrated dataset, cell-type-specific differentially expressed gene (DEG) analysis between AD and control groups revealed a relatively balanced distribution of DEG counts across cell types (Figure 1H), ranging from 140 DEGs (in inhibitory neurons) to 367 DEGs (in astrocytes), a variation within 3-fold difference (Table S1). In contrast, the original study by Mathys et al. reported a much greater disparity, from only 28 DEGs (in OPCs) and up to 756 DEGs (in excitatory neurons), more than a 20-fold difference²⁹ (Table S1). The increased sample size achieved through multi-study integration likely improves the ability to identify and analyze underrepresented cell types, such as OPCs and microglia, by capturing shared biological signals that might be weak or obscured within individual datasets.^{45,46} Thus, this integrated analysis improved consistency in DEG detection efficiency across cell types, enabling robust and unbiased downstream drug target screening using these cell-type-specific DEG profiles as inputs.

Notably, the cell-type variability in AD signatures was observed through distinct DEG identities, with significant proportions of DEGs unique to individual cell types (Figure 1H; Table S2). Although others were shared across cell types, opposite regulatory patterns in AD were displayed (Figure 1I). For example, the *APOE* gene, a major genetic risk factor for AD,⁴⁷ was significantly upregulated in microglia but downregulated in astrocytes and OPCs in AD samples. Additionally, gene set enrichment analysis showed that AD-enriched Kyoto Encyclopedia of Genes and Genomes (KEGG) pathways and Gene Ontology (GO) terms ex-

hibited extensive cell-type heterogeneity, with most pathways and terms unique to individual cell types (Figures 1J and S1D; Table S2). Although pathways such as chemical carcinogenesis by reactive oxygen species and prion disease were enriched in AD across all six cell types, unique pathways delineated them into two clusters. The neuronal-centric cluster included excitatory and inhibitory neurons, with oligodendrocytes sharing estrogen signaling with excitatory neurons and cAMP signaling with inhibitory neurons. The glial-centric cluster was comprised of astrocytes, microglia, and OPCs, with astrocytes and OPCs sharing gap junction pathway enrichment and microglia and OPCs enriched for oxidative phosphorylation. These findings suggest that AD pathogenesis involves heterogeneous transcriptome-driven molecular alterations manifested in the discordant behaviors of multiple cell types. Effective treatment likely needs to correct malfunctions in multiple, if not all, cell types.

Computational drug repurposing pipeline predicts cell-type-specific therapeutic candidates

Using cell-type-specific AD profiles from our integrated analysis, we screened for network-correcting drug candidates by querying each profile against the Connectivity Map (CMap),³² a drug expression database generated with human cancer cell lines, via a computational pipeline that matches gene expression profiles of diseases and existing drugs⁴⁸ (Figure 2A). Most AD signatures overlapped with drug profile features, which were then used as inputs in the pattern-matching algorithm (Figure 2B). With a false discovery rate (FDR) < 0.05, we predicted 35 hits for excitatory neurons, 12 hits for inhibitory neurons, 8 hits for microglia, 33 hits for astrocytes, 4 hits for oligodendrocytes, and 29 hits for OPCs, with several drug hits shared across cell types (Figure 2C).

Notably, 25 repurposed drugs significantly reversed cell-type-specific AD profiles in multiple cell types (Figure 2D), indicating a multi-targeted potential with these drugs. These multi-cell-type drug candidates span various therapeutic classes, including cardiac glycosides (digitoxigenin, digoxigenin, and helveticoside), chemotherapeutic agents (methotrexate, irinotecan, etoposide, and mitoxantrone), aromatase inhibitor (letrozole), histone deacetylase inhibitors (trichostatinA, vorinostat, and scriptaid), immunosuppressants/mTOR inhibitor (sirolimus), anti-inflammatory/anti-tumor agent (15-delta prostaglandin J2), nonsteroidal

Figure 2. Computational drug repurposing pipeline predicts drug candidates reversing cell-type-specific transcriptomic signature profiles of AD

- (A) Schematic showing computational drug repurposing workflow.
 (B) Input DEG counts (filtered list by mapping AD versus control DEGs to existing gene probes in the CMap database) per cell type for the computational drug repurposing pipeline.
 (C) Disease-drug network depicting the connections between the six cell types and drug candidates that significantly reverse disease profiles within the respective cell types. Drug names in red denote drug candidates validated in humans using electronic medical records (EMR).
 (D) Heatmap showing drug candidates that significantly reverse AD profiles in more than one cell type. Black frames label drugs validated in humans using EMR, and red asterisks label drugs selected for validation *in vivo* using a mouse model of AD.
 (E) AD prevalence assessment in drug-exposed individuals, only drugs with significant reduced AD risks are shown.
 (F) Schematic illustration demonstrating the rationale of prioritizing letrozole and irinotecan as potential combination therapy for further validation in a mouse model of AD.
 (G) Heatmaps of each cell-type-specific AD transcriptomic signature profiles, rank-ordered genes from the most upregulated to the most downregulated and color coded by log fold changes, in comparisons with the gene probe ranks by letrozole or irinotecan treatments, colored by corresponding fold change ranks in the CMap.

See also Figure S2 and Table S3.

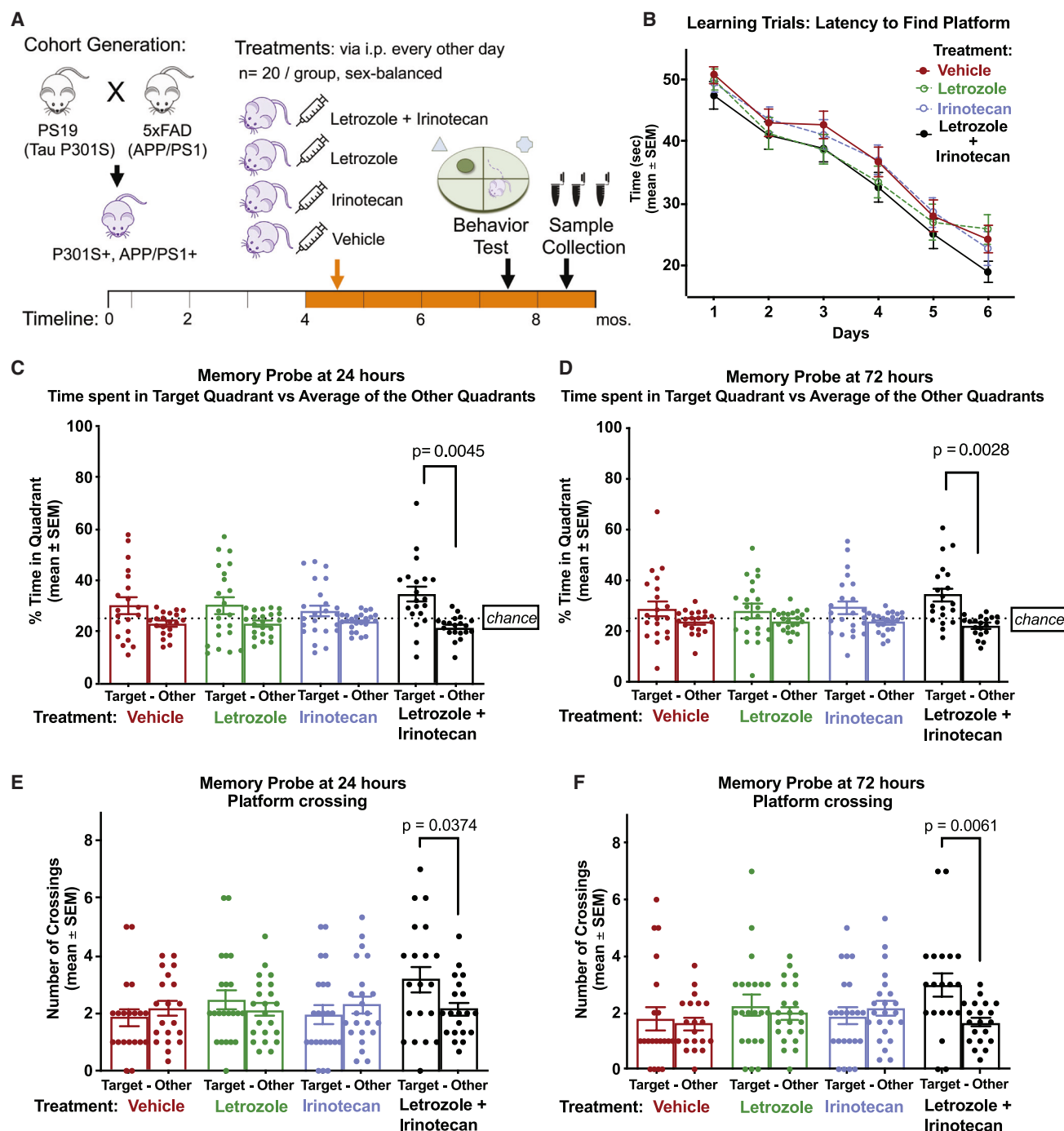


Figure 3. Combination treatment with letrozole and irinotecan rescues AD-like memory impairments in aged 5×FAD/PS19 mice with both Aβ and tau pathologies

(A) The schematic of treatment cohort design and experimental timeline (created with [BioRender.com](#)). Four treatment groups (n = 20 mice per group, both sexes), including treatment with vehicle, letrozole (1 mg/kg), irinotecan (10 mg/kg), or a combination of both drugs every other day via i.p. injection.

(B) Escape latency during hidden platform training days 1–6 in a Morris water maze (MWM) test did not differ statistically between groups. One-way repeated-measures ANOVA test was applied to compare vehicle group with treatment groups.

(C and D) Memory probes of percent time spent in target quadrant versus average of other quadrants demonstrated a significant preference of the target quadrant solely by mice with combination treatment at 24 h (C) and 72 h (D) after removing the hidden platform.

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anti-inflammatory drug (fenbufen), antifungal agent (ciclopirox), antibiotics (monensin and ionomycin), antiepileptic (valproic acid), antipsychotic (haloperidol), and several experimental compounds with undefined pharmacological classifications (LY-294002, CP-320650-01, CP-690334-01, syrosingopine, hycanthone, and 5155877).⁴⁹

Validation of candidate drug effects in humans using real-world evidence derived from electronic medical records

We aimed to validate the effects of the repurposed drug candidates in humans using real-world data. A distinct advantage of existing pharmaceutical agents is their potential for population-level analysis using real-world patient data to explore associations between drug use and AD outcomes. We explored the University of California (UC)-wide electronic medical records (EMR) database, which includes clinical records of more than 10 million individuals across six university health centers in California. We identified 1.4 million individuals aged 65 or older within the UC-wide EMR database, providing a substantial sampling pool for the population-based analysis.

Focusing on the 25 multi-cell-type candidate drugs, we found usage records for 10 of these drugs, as only a subset of the drugs from CMap is FDA approved and prescribed (Table S3). Relative risk was analyzed by comparing the occurrence of AD diagnosis between drug-exposed individuals and a propensity-matched non-exposed control cohort, balanced for key variables such as age, sex, race, ethnicity, comorbidities, and original indications (STAR Methods). Of these ten drugs, five (letrozole, irinotecan, methotrexate, ciclopirox, and sirolimus) were associated with a significantly reduced risk of AD compared with matched controls (Figure 2E), suggesting potential protective effects against AD. Although three drugs (etoposide, trifluoperazine, and vorinostat) showed reduced risk, statistical significance was not achieved due to insufficient patient representations (Table S3). Lastly, two drugs (valproic acid and haloperidol), both used for neurological conditions, showed higher relative risk for AD.

Prioritization of letrozole and irinotecan as combination therapy for AD based on human-derived evidence

We hypothesized that combination therapy targeting both neuronal and glial transcriptomic profiles might more effectively alleviate AD pathologies. Among five drug candidates showing significant AD risk reduction in UC-wide EMR, we prioritized letrozole for its predicted reversal effects in excitatory and inhibitory neurons and irinotecan for its effects on the glial-centric cluster, including astrocytes, microglia, and OPCs (Figure 2F). Therefore, a combination of letrozole and irinotecan potentially targets five cell types in AD.

The transcriptomic signatures in CMap were generated following a 6-h treatment at 14 μ M letrozole in the prostate can-

cer cell line PC3 and 100 μ M irinotecan in the breast cancer epithelial cell line MCF7, representing more acute drug responses in human cell lines.³² Letrozole reversed AD-associated gene expression in neurons, with upregulated genes shifting downward and downregulated genes shifting upward (Figure 2G). Reversal scores, where higher absolute values indicate a stronger reversal effect, were -0.37 in excitatory and -0.61 in inhibitory neurons. Irinotecan demonstrated similar reversal effects in glial cells, with reversal scores of -0.29 , -0.31 , and -0.45 for astrocytes, microglia, and OPCs, respectively (Figure 2G).

Letrozole, an aromatase inhibitor primarily prescribed for breast cancer treatment and occasionally for male infertility,^{50,51} had a relative risk ratio of 0.466 for AD in UC-wide EMR. Despite the female-skewed cohort (approximately 33 females per male), AD prevalence was comparable between sexes, with a relative risk score of 0.431 in females and 0.444 in males in the sex-stratified analysis (Table S3). However, the small sample size in males limits statistical power, rendering the effect of letrozole on AD outcome in males inconclusive based on EMR analysis. Irinotecan, a DNA topoisomerase I inhibitor used for colorectal and pancreatic cancer treatment,²⁸ showed a relative risk ratio of 0.195 for AD. Covariate balance between drug-exposed and matched controls was achieved through the propensity-matching algorithm^{52,53} with absolute standardized mean difference < 0.1 for all covariates, demonstrating adequate balance after matching (Figure S2). The age at first exposure ranged from 50 to 90+ years in both the letrozole- and irinotecan-exposed groups, with an average exposure age of 70.80 years for letrozole and 70.58 years for irinotecan. Among unexposed controls, the average age at AD diagnosis was 77.34 years in the letrozole-matched group and 72.78 years in the irinotecan-matched group, suggesting that drug exposure may occur several years before the onset of AD symptoms.

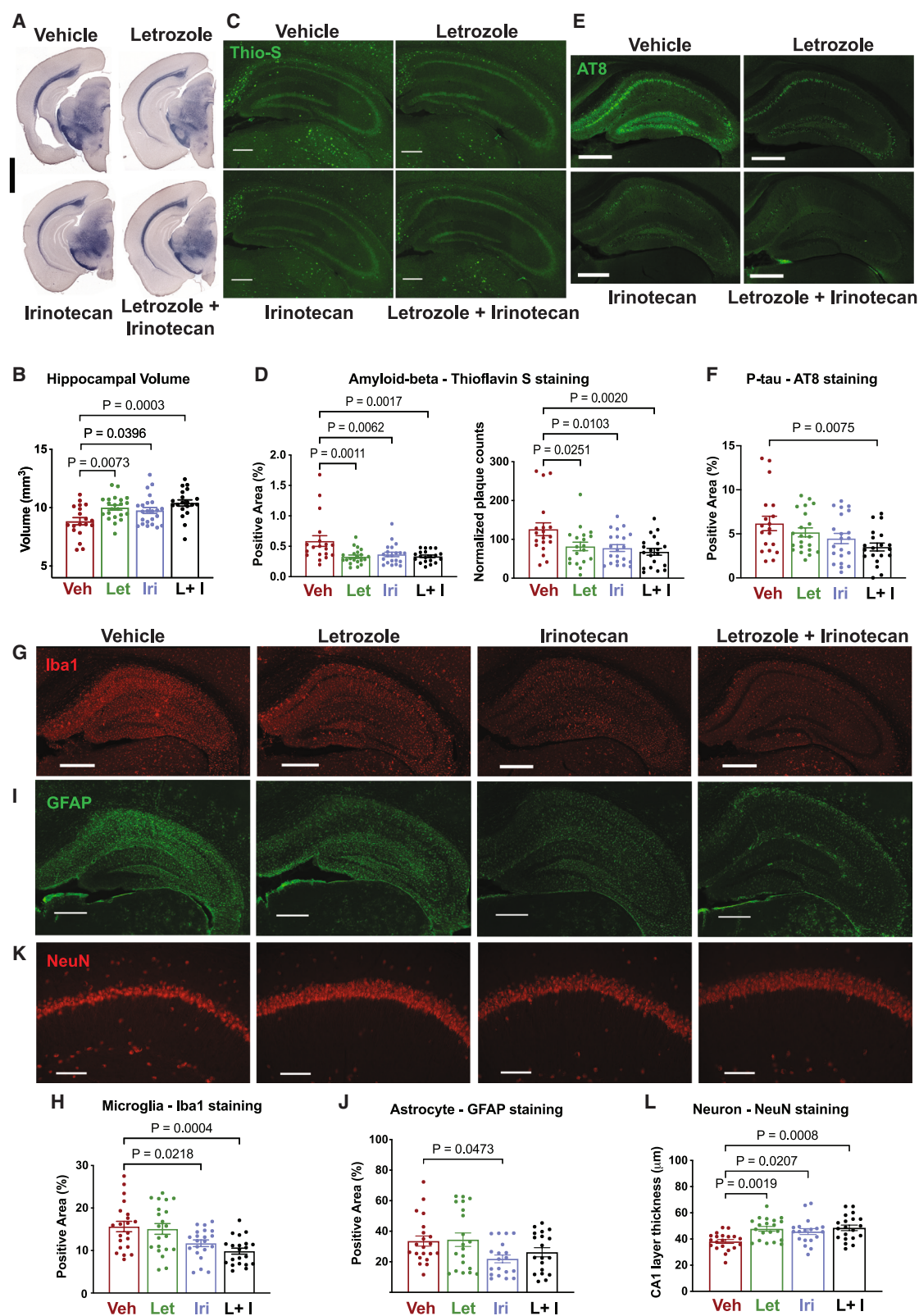
The combination treatment with letrozole and irinotecan rescues both short-term and long-term spatial memory in a mouse model of AD

To experimentally test the efficacy of the combination therapy with letrozole and irinotecan, we conducted dosing experiments in an AD mouse model ($5\times$ FAD³³ $\times$ PS19³⁴) expressing mutant human APP/PS1 and tau, which recapitulates many AD-related phenotypes, including amyloid plaque formation, tau tangles, and gliosis, with an early and aggressive pathology onset^{35,54} (Figure 3A). A sex-balanced double transgenic cohort was evenly divided into four groups ($n = 20$, including both sexes) and each treated with either vehicle, letrozole (at 1 mg/kg), irinotecan (at 10 mg/kg), or the combination of both drugs every other day via intraperitoneal (i.p.) injection for 3 months. To mitigate potential toxic effects associated with long-term dosing and better replicate human drug exposure levels, we selected

(E and F) Memory probes measuring number of platform-location crossings in the target quadrant versus average of the other quadrants. Significantly more crossings in the target quadrant (where the platform used to be) were only observed in the combination-treatment group at 24 h (E) and 72 h (D) after the platform was removed.

For (C)–(F), ordinary one-way ANOVA test was performed between target and other quadrants for each group, and p value was adjusted with Bonferroni multiple-comparisons testing. Data are represented as mean $\pm$ SEM.

See also Figures S3–S5 and Table S4.



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concentrations within tolerable dosing ranges reported in mice that are equivalent to human dosage yet significantly lower than those used in CMap (STAR Methods). This design ensures that dosing remains clinically relevant and closely mimics human responses observed in the EMR analysis.^{27,28,55–60}

Spatial learning and memory performance were evaluated using the Morris water maze (MWM) test.⁶¹ Although no significant difference was observed in hidden platform training trials across 6 days between each treatment and vehicle group (Figure 3B), the memory test in probe trials revealed that only the combination-treatment group exhibited a statistically significant preference for the target quadrant at both 24 and 72 h post training (Figures 3C and 3D), suggesting a rescue of both short-term and long-term memory deficits by the combination treatment. Furthermore, only the combination-treatment group demonstrated significantly better location recall by making more frequent crossings to the platform location at both time points after platform removal (Figures 3E and 3F). Mice in the combination-treatment group also significantly outperformed the vehicle-treated mice in latency to the first platform crossing during the 72-h probe trial (Figures S3A and S3B). Average swim speeds were similar across all groups, indicating that the observed behavioral differences were not confounded by visual or motor impairments (Figures S3C, S3D, and S4A–S4D).

Additionally, sex differences were observed in the dosing experiment, with significantly improved learning performance only observed in combination-treated males compared with vehicle-treated males and not in females (Figures S4E and S4F). Memory rescue by the single drug treatments was also evident in males during the 24- and 72-h probe trials but not in females (Figures S4G–S4N). Taken together, AD-related behavioral assessments demonstrated that only the combination therapy with letrozole and irinotecan effectively rescued memory deficits in an AD mouse model with both amyloid and tau pathologies. The combination therapy also improved spatial learning in males only. Notably, the combination treatment consistently outperformed single-drug treatments, which exhibited markedly less pronounced efficacies across multiple probes.

The combination treatment with letrozole and irinotecan rescues AD pathologies in a mouse model of AD

To determine whether the single or combination drug treatments can also rescue AD-related pathologies, we morphologically assessed the cohort at 9 months of age (4 months post treatment). According to previous literature, this AD mouse model develops extensive neurodegeneration, A β deposits, hyperphosphorylated tau (p-tau) accumulation, gliosis, and severe loss of CA1 neurons in the hippocampus at this age.^{35,54} We first evaluated neurodegeneration in the hippocampal region, as atrophy in this region is a hallmark of AD progression. Analyses of hippocampal volume showed rescue of atrophy in all treatment groups, with the most significant improvement in the combination-treatment group compared with the control group (Figures 4A and 4B). Thioflavin S (Thio-S) staining for A β pathology revealed significant reductions in all treatment groups compared with vehicle-treated controls, measured by the Thio-S-positive percent area and plaque counts normalized by hippocampal size (Figures 4C and 4D). P-tau pathology was assessed by immunofluorescent staining using the p-tau-specific AT8 antibody. Although all treatment groups showed a trend of reduced p-tau pathology compared with vehicle-treated controls, only the combination-treatment group had a statistically significant reduction in the p-tau coverage area of the hippocampus (Figures 4E and 4F).

Given the cell-type-precision therapeutic design, we investigated the effects of each treatment on gliosis and neuronal loss through immunostaining of specific cell types. As major contributors to neuroinflammation, microgliosis and astrogliosis were evaluated by quantifying the coverage area of cell-type-specific markers Iba1 and Glial Fibrillary Acidic Protein (GFAP), respectively. Mice in irinotecan and combination-treatment groups showed a significant reduction in the Iba1-positive percent area in the hippocampus, indicating alleviation of microgliosis (Figures 4G and 4H). Astrogliosis reduction was moderate, with significant reduction observed only in the irinotecan-treated group (Figures 4I and 4J). Neuronal layer thickness, assessed via NeuN staining, was employed to quantify neuronal

Figure 4. AD pathologies are significantly reduced in aged 5 $\times$ FAD/PS19 mice after drug treatments, with the strongest rescue in the combination-treatment group

(A) Representative images of the ventral hippocampus from 9-month-old 5 $\times$ FAD/PS19 mice with Sudan Black staining to enhance hippocampal visualization (Scale bar, 2 mm).

(B) Quantification of hippocampal volume in 9-month-old 5 $\times$ FAD/PS19 mice across treatment groups.

(C) Representative images of Thio-S staining in the hippocampus from 9-month-old 5 $\times$ FAD/PS19 mice (Scale bar, 500 μ m).

(D) Quantification of Thio-S-positive percent coverage area and plaque counts in the hippocampi of 9-month-old 5 $\times$ FAD/PS19 mice across treatment groups.

(E) Representative images of hippocampus from 9-month-old 5 $\times$ FAD/PS19 mice with immunostaining of phosphorylated-tau (p-tau) using AT8 monoclonal antibody (Scale bar, 500 μ m).

(F) Quantification of AT8-positive p-tau percent coverage area in 9-month-old 5 $\times$ FAD/PS19 mice across treatment groups.

(G) Representative images of microglia immunostaining with anti-Iba1 in the hippocampi of 9-month-old 5 $\times$ FAD/PS19 mice (Scale bar, 500 μ m).

(H) Quantification of the percent Iba1 coverage area in the hippocampi of 9-month-old 5 $\times$ FAD/PS19 mice across treatment groups.

(I) Representative images of astrocyte immunostaining with anti-GFAP in the hippocampi of 9-month-old 5 $\times$ FAD/PS19 mice (Scale bar, 500 μ m).

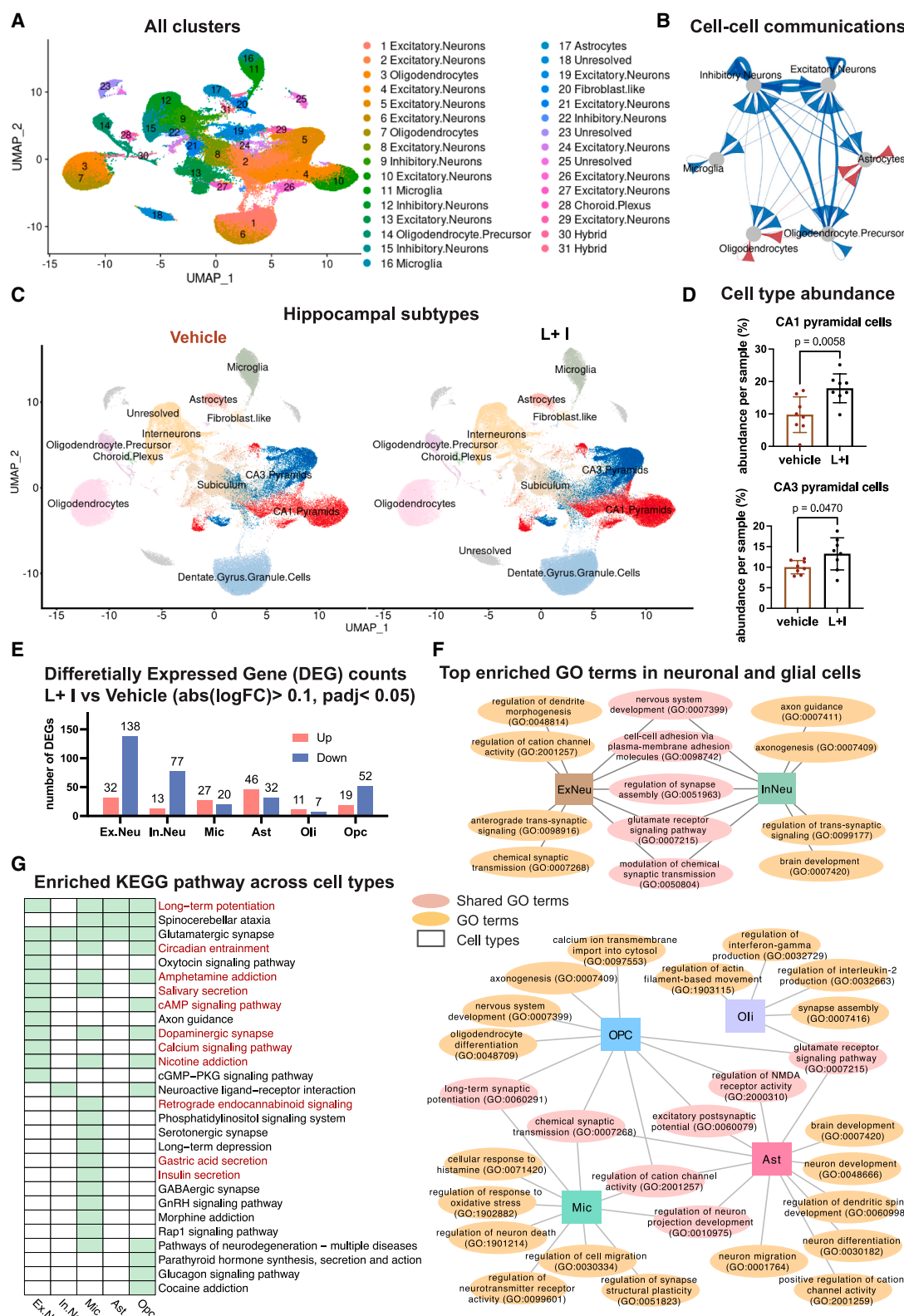
(J) Quantification of percent GFAP coverage area in hippocampi of 9-month-old 5 $\times$ FAD/PS19 mice.

(K) Representative images of CA1 neurons with neuronal marker NeuN immunostaining (Scale bar, 100 μ m).

(L) Quantification of the thickness of the CA1 neuronal cell layer of 9-month-old 5 $\times$ FAD/PS19 mice across treatment groups.

For (B), (D), (F), (H), (J), and (L), ordinary one-way ANOVA was applied between all treatment groups and vehicle control group, and *p* values were adjusted by Dunnett's multiple comparison test. For all experiments, *n* = 20 mice per group (both sexes). Data are represented as mean $\pm$ SEM.

See also Figures S3 and S5 and Table S4.



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loss,^{62,63} with a specific focus on the CA1 region of the hippocampus because previous studies suggest that CA1 volume correlates with AD severity.^{64,65} Significant rescue of neuronal loss was evident in the CA1 region (Figures 4K and 4L). The most significant rescues were observed in letrozole and combination-treatment groups. Interestingly, irinotecan-treated mice also showed moderate rescue in CA1 neuronal loss, likely attributable to the alleviation of gliosis that may help prevent neuronal loss.

To evaluate the relationship between cognitive performance and underlying neurodegenerative changes, correlation analysis was conducted between behavioral test metrics and pathological measurements. Significant correlations were observed between memory probes, percent time spent in the target quadrant, and pathology metrics, including NeuN (CA1) layer thickness, hippocampal volume, p-tau level, and A β plaque counts (Figures S3E–S3K). Notably, the combination-treatment group exhibited consistently stronger correlations compared with the vehicle-treated group, particularly between the percent of time spent in the target quadrant during memory probes and the thickness of NeuN-positive CA1 layers. These findings suggest that improvements in cognitive performance in the combination-treatment group were more closely associated with reductions in pathological markers, reinforcing the potential therapeutic synergy observed.

A two-way ANOVA test assessing variance contributions revealed that letrozole and irinotecan exert independent effects on multiple behavioral and neuropathological measures, with minimal evidence of interactive effect (Table S4). Letrozole significantly contributed to the increased 72-h target quadrant crossings, while irinotecan had a non-significant contribution to behavior metrics. Both drugs independently increased CA1 neuronal layer thickness, with a significant interaction suggesting a potential combined effect on neuronal integrity (Figure S5). Hippocampal volume, Thio-S+ plaque count, and AT8 coverage area were influenced by both drugs, but their combined effects were more additive than synergistic, with no significant interactive effect. Irinotecan significantly reduced neuroinflammation markers, including Iba1 (microglia) and GFAP (astrocytes), whereas letrozole had no impact on these measures. Overall, these findings suggest that letrozole primarily influences neuronal integrity measures, whereas irinotecan has a stronger anti-inflammatory effect. Although the combination treatment appears to yield enhanced therapeutic benefits,

mechanistic synergy between the two drugs remains unclear. Instead, they may exert their effects through distinct biological processes rather than directly reinforcing each other's mechanisms of action.

The combination treatment with letrozole and irinotecan promotes neuroprotective functional pathways in a cell-type-specific manner

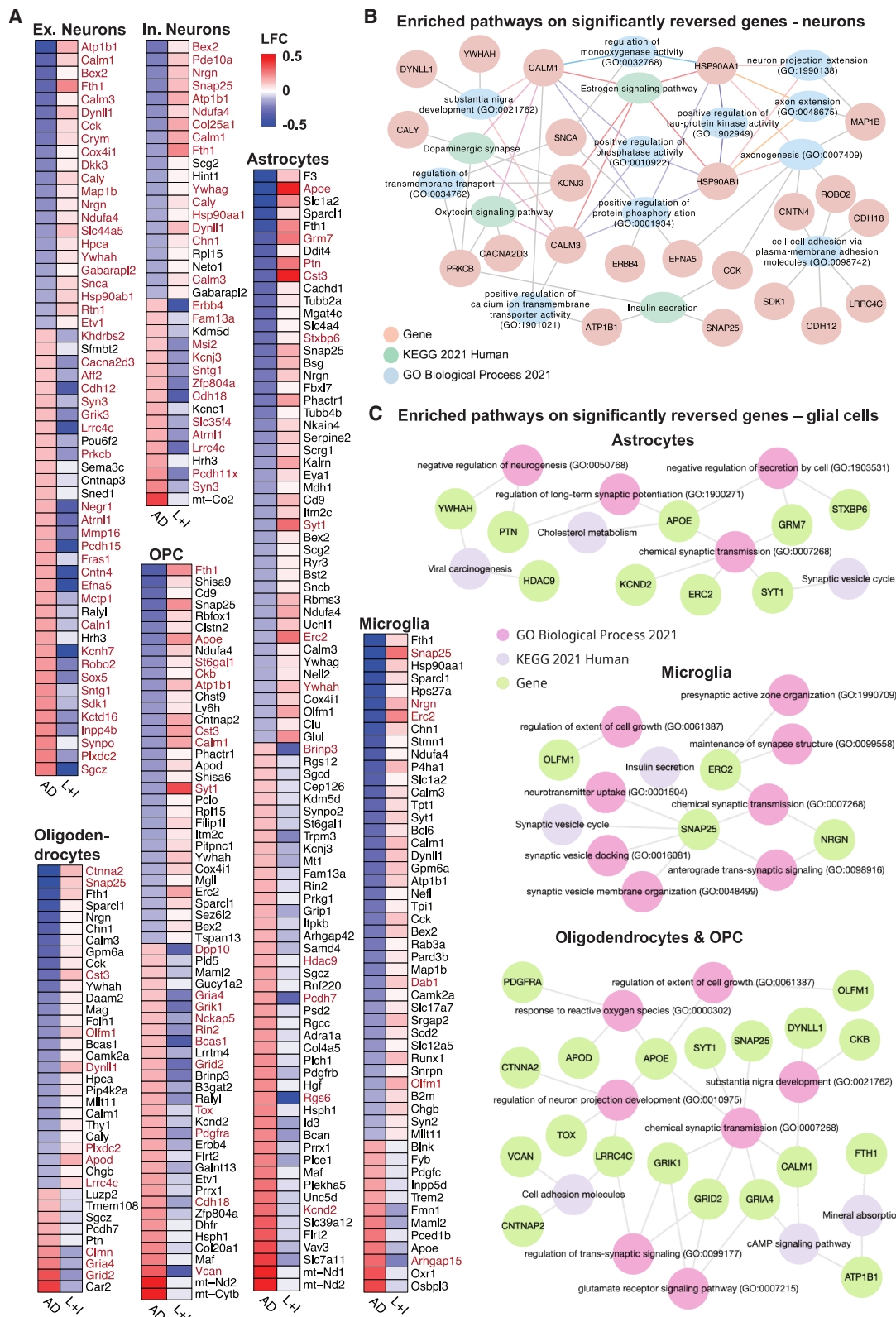
To investigate the cell-type-specific transcriptomic changes in response to the combination treatment, snRNA-seq was performed on dissected hippocampi obtained from mice on combination treatment of letrozole and irinotecan (L + I) or vehicle treatment ($n = 8$ per group, including both sexes). UMAP visualization revealed 31 distinct cell clusters, including clusters assigned to the six major cell types (Figure 5A). Additionally, L + I treatment reduced excessive CCC across multiple cell types (Figures 5B and S6A), including AD-increased communications (Figure 1G). This broad reduction could suggest that the treatment reduces the hyperactive and dysregulated neuronal-glia interactions in AD. For instance, inhibitory neuron-to-astrocyte CCC, which was elevated in AD, decreased with L + I treatment, suggesting that reducing astrocyte-driven inhibition may help restore cognitive function. Similarly, the treatment reduced excitatory neuron-to-OPC and oligodendrocyte CCC, potentially stabilizing excitatory neuron support.

To account for the granularity and selective vulnerabilities among neuronal subtypes, we subdivided the excitatory and inhibitory neurons into CA1 pyramidal cells, CA3 pyramidal cells, dentate granule cells (DGC), subiculum neurons, and interneurons based on expressions of hippocampal subregion marker genes. Notably, in the UMAP plot split by treatment groups, there is a discernible higher density of cells observed in the L + I-treated group compared with the vehicle-treated group within the pyramidal neuron clusters of the CA1 and CA3 regions (Figure 5C). This was quantified via cell type abundance analysis, revealing that the proportions of CA1 and CA3 pyramidal neurons in L + I-treated mice were significantly higher than in vehicle-treated mice (Figure 5D), consistent with our findings from pathological analysis. There were no significant differences in the abundances of other hippocampal cell types between L + I-treated and vehicle-treated groups (Figures S6B and S6C).

Differential gene expression analysis comparing L + I and vehicle treatments revealed varying counts and compositions

Figure 5. Single-nucleus RNA sequencing analysis in aged 5 $\times$ FAD/PS19 mice across treatment groups

- (A) UMAP plot of all 31 distinct cell clusters in the hippocampi of 9-month-old mice from combination-treatment (L + I) and vehicle-treatment groups.
- (B) Projection plots for CCC measured by expression of receptor and ligand pairs and represented as arrows connecting two cell types. Red connections indicate increased communication and blue indicate decreased communications comparing L + I-treated versus vehicle-treated groups. Arrows point to the directions from sender to receiver of the communication.
- (C) UMAP plots labeled by hippocampal-specific cell types and split by treatments to highlight difference in distribution density for neuronal subtypes between treatments.
- (D) Scattered bar plots of cell type abundance (percentage) for CA1 and CA3 pyramidal neurons (unpaired two-sided t test; data are represented as mean $\pm$ SEM), comparing L + I-treated and vehicle-treated mice ($n = 8$ per group, including both sexes).
- (E) Differentially expressed gene (DEG) counts of L + I-treated versus vehicle-treated mice in 6 major brain cell types. Counts of DEGs per cell type do not correlate with cell counts.
- (F) Top L + I-treatment-enriched Gene Ontology terms across six cell types.
- (G) Heatmap illustration of L + I-treatment-perturbed KEGG pathways across 6 cell types. Many pathways overlap with AD-enriched KEGG pathways (see Figure 1J) and are highlighted in red.
- See also Figure S6 and Table S5.



(legend on next page)

of DEGs ($\text{abs}(\log\text{FC}) > 0.1$, $\text{padj} < 0.05$) across cell types, with the fewest DEGs observed in oligodendrocytes, suggesting that they were least affected (Figures 5E and S6D; Table S5). Gene set enrichment analysis of cell-type-specific DEGs revealed enrichments in disease-relevant functional pathways (Figure 5F). Notably, the treatment-enriched pathways related to nervous system development and synaptic activities in neurons. For example, in excitatory neurons, the dendrite morphogenesis pathway was enriched, while the axonogenesis pathway was enriched in inhibitory neurons. In glial cells, the oligodendrocyte differentiation pathway was enriched in OPCs, and actin-filament-based movement and synapse assembly pathways in oligodendrocytes. Essential microglial functions like pathways of response to oxidative stress and histamine, regulation of neuron death and synaptic plasticity, and neuron projection development were enriched, as were neuron development and regulation of synaptic activity pathways in astrocytes. KEGG pathway analysis revealed that L + I treatment perturbed AD-associated pathways from our integrated analysis (Figure 1J), such as long-term potentiation, circadian entrainment, cAMP signaling, and calcium signaling (Figure 5G; Table S5), suggesting that the combination treatment promotes neuroprotective functional pathways in a cell-type-specific manner.

The combination treatment with letrozole and irinotecan effectively reverses multiple cell-type-specific transcriptomic signatures of AD

Lastly, we explored combination-treatment-reversed transcriptomic signatures of AD to elucidate potential molecular and cellular mechanisms underlying the observed benefits of the L + I treatment. We mapped human AD signatures to homologous mouse genes (Figure S6E). AD signatures from human-integrated analysis were regulated in opposite directions by L + I treatment ($\text{abs}[\log\text{FC}] > 0.01$) in a cell-type-specific manner. In particular, 56 AD signatures in excitatory neurons, 36 AD signatures in inhibitory neurons, 90 AD signatures in astrocytes, 52 AD signatures in microglia, 37 AD signatures in oligodendrocytes, and 64 AD signatures in OPCs were inversely regulated by L + I treatment in mice (Figure 6A). With a padj value < 0.05 cutoff for differential gene expressions in L + I treatment compared with vehicle treatment, 49 and 27 AD signatures were significantly reversed by L + I treatment in excitatory and inhibitory neurons, respectively, with 13 shared between them. In glial cells, fewer reversed gene expressions reached statistical significance: 13 in astrocytes, 6 in microglia, 11 in oligodendrocytes, and 19 in OPCs. However, they unveiled intriguing patterns. For instance, *APOE*, identified as a DEG in the human-integrated

analysis, was upregulated in microglia and downregulated in astrocytes and OPCs (Figure 1I). The expression patterns of *ApoE* in the tested AD mouse model were reversed by L + I treatment across all three cell types (Figure 6A), although statistical significance was only achieved in astrocytes and OPCs, possibly due to the presence of a small but diverse microglial population in the sequenced cohort.

We then conducted gene set enrichment analysis on significantly reversed genes to identify potential mechanistic targets. Combining reversed genes in excitatory and inhibitory neurons revealed associations with the estrogen-signaling pathway, among others (Figure 6B). Reversed genes within this pathway are also involved in regulation of protein kinase and phosphatase activity, as well as tau kinase activity, highlighting potential mechanisms by which letrozole may modulate neuronal dysfunction in AD. Furthermore, certain reversed genes are associated with synaptic activities and neuron projections, potentially contributing to the rescue of neurodegeneration noted in the pathological analysis. In astrocytes, reversed genes are associated with the regulation of long-term synaptic potentiation, chemical synaptic transmission, and cholesterol metabolism (Figure 6C). In microglia, reversed genes are associated with pathways regulating synapses and cell growth (Figure 6C). Additionally, we combined reversed genes from oligodendrocytes and OPCs, revealing associations with pathways related to cell growth regulation, response to reactive oxygen species, and neuron projection regulation (Figure 6C). These findings provide a transcriptomic foundation, supporting the network correction concept that L + I combination treatment rescues AD-related behavioral and pathological deficits by rectifying complex dysregulated gene networks across multiple disease-relevant cell types.

DISCUSSION

Despite the initiation of over 2,700 clinical trials in the past two decades, primarily targeting single-core pathologies, only three therapies targeting A β plaques have recently received FDA approval.⁶ This includes Aducanumab, which was discontinued in 2024.⁶⁶ The limited success of single-target approaches underscores the need for strategies addressing AD's heterogeneous pathologies across various cell types, such as synaptic loss, neuronal death, neuroinflammation, cholesterol homeostasis alteration, energy metabolism deficit, dysfunctional phagocytosis, and demyelination.^{14,15} Therefore, we designed a cell-type-directed combination therapy to precisely correct dysregulated gene networks in both

Figure 6. Combination treatment with letrozole and irinotecan reverses cell-type-specific transcriptomic signatures of AD

(A) Comparison of cell-type-specific transcriptomic signature profiles of AD in humans with gene expression changes in combination-treatment (L + I) versus vehicle-treatment groups of mice. Only treatment-reversed genes from AD profiles are included ($\text{abs}(\log\text{FC}) > 0.01$ in L + I-treated versus vehicle-treated mice), and those with FDR-adjusted $p < 0.05$ are labeled in red. Heatmap colors indicate directions and magnitude of gene expression changes, with downregulations in blue and upregulations in red.

(B) Gene set enrichment analysis on L + I-treatment-reversed genes, which reached statistical significance in excitatory and inhibitory neurons, revealed reversal of multiple AD relevant functional pathways.

(C) Gene set enrichment analysis on L + I-treatment-reversed genes, which reached statistical significance in astrocytes, microglia, or oligodendrocytes and OPCs, demonstrated reversal of glial-cell-specific AD gene signatures and functional pathways.

See also Figure S6 and Table S6.

neuronal and glial cells. This approach employs a network-correction strategy to restore essential cellular functions by simultaneously targeting multiple AD-related pathways. By reversing the expression of hundreds of disease signature genes within distinct cell types, this strategy offers a comprehensive and synergistic means of addressing complex mechanisms underlying AD.

Although systems biology approaches and single-cell data mining have been employed for computational AD drug repurposing,^{67,68} these attempts often yield extensive lists of candidates without comprehensive validation or prioritization, thereby limiting clinical translation. To address this challenge, we leveraged real-world evidence from the UC-wide EMR to examine the prevalence of AD diagnosis in patients after drug exposure. Among the predicted drug candidates with sufficient patient representation, the majority were associated with decreased relative risk for AD, indicating potential therapeutic benefits in humans. Based on their intended cell-type-specific targets, we prioritized letrozole (targeting neurons) and irinotecan (targeting glia) as a combination therapy from the EMR-validated drugs. Both drugs also demonstrate the ability to cross the blood-brain barrier,^{56,59} ensuring delivery to the brain for therapeutic activity. Epidemiological evidence further supports this prioritization, with studies showing reduced dementia risk among breast cancer patients treated with letrozole⁶⁹ and decreased AD risk among colorectal cancer survivors treated with irinotecan.⁷⁰ Together, these findings provide strong clinical and mechanistic support for the potential efficacy of letrozole and irinotecan as a combination therapy for AD.

Unlike randomized controlled trials, observational analyses using EMR present challenges, particularly the potential for systematic differences between drug-exposed and unexposed individuals, contributed by confounding covariates. In our study, both letrozole and irinotecan are primarily used in cancer patients, a population previously associated with lower AD incidence.⁷¹ To mitigate potential bias, we applied propensity score matching (PSM)⁵² to identify unexposed control groups balanced for covariates, including age, sex, race, ethnicity, location, AD-associated comorbidities (e.g., hypertension), and original indications (breast cancer and infertility for letrozole; colorectal and pancreatic cancer for irinotecan). We restricted the analysis to individuals aged 65 and older and implemented a 1:2 matching scheme, where each drug-exposed individual was matched with two unexposed controls. After matching, balance was achieved between the groups for the covariates (Figure S2). At the time of analysis, AD prevalence among all individuals aged 65 or older with a cancer diagnosis in the UC-wide EMR was 2.4% (2,852 AD cases/118,103 cancer cases), lower than the general population rate of 4% reported in the National Health Statistics Reports.⁷² In our matched control groups, the prevalence of AD was 2.18% for letrozole and 1.83% for irinotecan (Table S3), both lower than the national average and consistent with rates observed among cancer patients in the UC-wide EMR. These findings support the effectiveness of our matching strategy in minimizing bias and ensuring robust comparisons.

For *in vivo* dosing validation, we crossed two widely used AD mouse models, 5×FAD³³ (with A β pathology) and PS19³⁴ (with

tau pathology). This approach was designed to better recapitulate AD patient profiles identified in our integrative human snRNA-seq analysis, where cases and controls were standardized across studies using CERAD (A β pathology) and Braak (tau pathology) scores. Both mouse models exhibit microgliosis and astrogliosis,^{33,34,73,74} ensuring the inclusion of glial cell pathology in our study. Although the 5×FAD/PS19 cross has been characterized in only a few studies,^{35,54} each mouse line is extensively used for studying AD-related neurodegeneration.^{73–77} To better recapitulate human drug responses in mice, we selected drug concentrations comparable to those used in humans and within reported mouse studies,^{27,28,57,59} but significantly lower than the concentrations used in CMap for the initial screening. The mouse study captures long-term effects over several months, whereas CMap transcriptomic responses reflect an acute response at just 6 h. Potential deviations in transcriptomic profiles may therefore arise due to dose- and duration-dependent effects.

Treatment in mice revealed significant rescue of memory impairment exclusively in the combination-treatment group. The combination therapy demonstrated robust alleviation across all pathology markers except astrogliosis. In contrast, single-drug treatments only partially rescued subsets of the investigated metrics, highlighting the superior therapeutic effectiveness of the combination approach. Notably, only the combination treatment significantly reduced p-tau pathology, a key predictor of cognitive decline in AD patients according to clinical studies,^{78,79} potentially accounting for the statistically significant memory improvement observed exclusively with the combination therapy. These findings underscore the importance of targeting both neuronal and glial cell pathologies to achieve a more comprehensive and effective treatment outcome.

The network-correcting effects of the combination therapy were investigated through snRNA-seq analysis of hippocampal samples from combination-treated and vehicle-treated mice. In addition to rescuing memory deficits and pathologies, the combination treatment reversed the expression of AD signature genes enriched across diverse AD-related pathways. In neurons, these included synaptic activity, axonogenesis, neuronal projections, and protein phosphorylation regulation; in astrocytes, cholesterol metabolism and long-term synaptic potentiation; in microglia, synaptic organization and cell growth regulation; and in oligodendrocytes and OPCs, regulation of neuronal projections and synaptic signaling. These findings support the hypothesis that a targeted, multifaceted therapeutic approach confers AD-specific benefits by rectifying multiple pathological processes in a cell-type-specific manner. Further investigation is required to delineate the precise mechanisms and key pathways modulated by the combination therapy.

Our snRNA-seq analysis elucidates the transcriptomic alterations induced by combined L + I treatment in a mouse model. However, it primarily reflects long-term systemic responses, making it difficult to distinguish between immediate AD signature reversal induced by the drugs and system-wide adaptation mechanisms over time. CMap profile analysis suggests that letrozole and irinotecan modulate both distinct and overlapping

pathways (Figure S6F). Both drugs perturb protein-binding and prion disease pathways, suggesting potential additive effects. Additionally, letrozole primarily influences synaptic and metabolic activity, while irinotecan affects glial cell homeostasis and metabolic processes, suggesting potential complementary beneficial effects across cell types. Although CMap provides a more direct measure of drug-induced AD signature reversal for individual drugs, it differs from the mouse study in key aspects, including cell type, dosage, and treatment duration. Future studies using cultured brain cell lines treated at human-relevant dosages over both short- and long-term time points will clarify the cell-type-specific contributions of each drug to AD signature reversal.

The combination of letrozole and irinotecan has been shown to reduce amyloid plaques and one form of p-tau in our mouse study, which may be a main driver of downstream beneficial effects. Pathway enrichment analysis of letrozole- and irinotecan-reversed genes in CMap suggests that both drugs regulate protein modulations but, more importantly, influence broader network-level changes across multiple pathological pathways (Figure S6F; Table S6). These include neuroinflammation (such as MAPK signaling in microglia), synaptic dysfunction (including neurotransmitter receptor activity in excitatory neurons and cell communication in inhibitory neurons), and metabolic alterations (such as mitochondrial ATP synthesis-coupled electron transport in excitatory neurons and amino acid import across the plasma membrane in astrocytes). Additionally, pathways related to nervous system development in excitatory neurons, inhibitory neurons, and OPCs, as well as cellular homeostasis in microglia and OPCs, were predicted to be perturbed by both drugs independently (Figure S6F) and observed to be enriched following combination treatment in mice (Figure 5F). These findings highlight the complex interplay between direct modulation of AD-related gene expressions and reductions in amyloid and pathological tau. Although the primary mechanisms driving the observed memory and pathological benefits remain unclear, both are likely to contribute to the therapeutic effects.

In summary, this study introduces a transformative approach to AD drug discovery by focusing on cell-type-specific interventions and predicting multi-target drug combinations based on human-derived data. Utilizing extensive omic data and EMR from millions of diverse patients, our methodology provides a robust framework for identifying effective therapeutic agents through real-world evidence. Preclinical validation in an AD mouse model demonstrated that the proposed drug combination significantly improved memory deficits and alleviated AD-related pathologies, whereas the single drugs alone showed limited or no efficacy, highlighting the superior therapeutic potential of the combination treatment. Our finding that the treatment reversed AD-specific gene networks supports the multi-cellular network correction concept of this strategy and enables further investigation of its therapeutic mechanisms. Besides highlighting the translational potential of the proposed drug combination, this study also exemplifies the promise of a cell-type-directed, network-correcting treatment strategy. It paves the way for precision medicine leveraging AI and large-scale patient-level data. Integrating

advanced computation with evolutionary-scale multi-omic, clinical, and drug perturbation databases, future efforts can develop AI systems to design tailored treatments for patients based on molecular signatures and clinical profiles of specific subpopulations or individuals.

Limitations of the study

Despite these promising findings, several limitations of this study should be noted. First, the CMap database was measured in cancer cells and may not accurately reflect brain tissue profiles; it was used to screen drug candidates and requires validations in neurological models. Relevant drug perturbation databases on brain cell types are currently unavailable. Second, although necessary, validation in animal models may not fully recapitulate human biology. Significant sex differences to treatment were observed in behavior tests, with male mice generally responding better behaviorally to all treatments (Figure S4). Sex differences in AD are well documented and the impact of sex hormones in AD remain inconclusive.^{80,81} As a hormone modulator, letrozole might contribute to this sex difference. In the UC-wide EMR sex-stratified analysis, relative risk scores for AD were similar between females and males following letrozole exposure. However, the analysis remains inconclusive due to the small number of male letrozole users. The observed sex differences may be specific to the mouse models, as both 5×FAD and PS19 have shown strong sex differences in prior studies.^{74,82} Additionally, EMR analysis presents potential caveats, as data tend to be sparse and are not collected with specific research in mind. Despite efforts to propensity-match, the control populations may not exactly resemble the drug-exposed populations if relevant diagnoses are missing. Therefore, rather than using relative risk scores as a measure of drug effectiveness, we relied on EMR results to prioritize drug candidates for further experimental validation.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources or reagents should be directed to, and will be fulfilled by, the lead contact, Marina Sirota (marina.sirota@ucsf.edu).

Materials availability

Materials generated during this study are accessible via reasonable request to the corresponding author's labs.

Data and code availability

The Mathys et al. dataset is accessible at [Synapse:syn18485175](https://synapse.org/syn18485175). The Zhou et al. dataset is accessible at [Synapse: syn21125841](https://synapse.org/syn21125841). The Lau et al. dataset is available at GEO: GSE157827.

Our integrated human snRNA-seq dataset and the snRNA-seq data in combination and vehicle-treated mice were uploaded to Mendeley Data: <https://doi.org/10.17632/f5kx9k264n.1>.

Code for transcriptomic data analysis and the multi-study integration using standard packages as described in the [STAR Methods](#) is available on Github at <https://github.com/YaqiaoLi8/CombinationLI>. Code for computational drug repurposing pipeline associated with the current submission is available at <https://doi.org/10.1053/j.gastro.2017.02.039>.

The UCSF EMR and code associated with the deidentified data analysis are available to UCSF-affiliated individuals, who can contact

UCSF's Clinical and Translational Science Institute (CTSI) (ctsi@ucsf.edu) or the UCSF's Information Commons team for more information (info.common@ucsf.edu). UC-wide EMR is only available to UC researchers who have completed analyses in their respective UC first and have provided justification for scaling their analyses across UC health centers (more details at <https://www.ucop.edu/uchealth/departments/center-for-data-driven-insights-and-innovations-cdi2.html> or by contacting healthdata@ucop.edu).

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AUTHOR CONTRIBUTIONS

Y.L., Y. Huang, and M.S. designed the study. Y.L. performed the majority of the studies and data analyses. C.P.S. performed cell-cell-communication analysis. J.B. aided immunohistochemical studies. M.X. facilitated treatment dosing experiments. Y. Hao and A.A. prepared samples for snRNA-seq. E.D., Y.Y.C., and J.H. performed animal behavior tests. S.Y.Y. and A.Z. aided in cohort tissue collection. A.R., S.W., A.T., J.S., X.T., and I.L. helped in developing and optimizing experimental procedures and conditions. T.O. and M.J. K. provided advice and guidance on the study. Y. Huang and M.S. supervised the study. Y.L., Y. Huang, and M.S. wrote the manuscript. All authors have read and approved the manuscript.

DECLARATION OF INTERESTS

Y. Huang is a co-founder and scientific advisory board member of GABAeron Inc. Y.L., Y. Huang, and M.S. are co-inventors on a patent application based on this study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse anti-AT8	Invitrogen	Cat# MN1020; RRID: AB_223647
Mouse anti-GFAP	Millipore Sigma	Cat# MAB3402; RRID: AB_94844
Rabbit anti-IBA1	Wako	Cat# 019-19741; RRID: AB_839504
Guinea pig anti-NeuN	Millipore Sigma	Cat# ABN90; RRID: AB_11205592
Donkey anti-mouse 488nm	Abcam	Cat# A-21202; RRID:AB_141607
Donkey anti-rabbit 594nm	Abcam	Cat#A-21207; RRID:AB_141637
Donkey anti-guinea pig 594nm	Jackson Immuno	Code: 706-585-148 RRID:AB_2340474
Chemicals, peptides, and recombinant proteins		
Mouse-on-Mouse Blocking Reagent	Vector Labs	Cat# MKB-2213
Normal Donkey Serum	Jackson Labs	Cat# 017-000-121
Triton-X	Millipore Sigma	Cat# T8787-250mL
Tween-20	Millipore Sigma	Cat# P2287-500mL
PBS	Thermo Fisher Scientific	Cat# 14190250
Avertin (2,2,2-Tribromoethanol)	Millipore Sigma	Cat# T48402
ProLong Gold Antifade Mountant	Thermo Fisher Scientific	Cat# P36930
Sudan Black	Sigma	Cat# 199664-25G
Thioflavin S	Millipore Sigma	Cat# T1892-25G
Irinotecan hydrochloride 5G	Fisher Scientific Warehouse	Cas #100286-90-6; Cat# 50-148-325
Letrozole	Fisher Scientific	Cat# L02481G
Saline solution 0.9%	Thermo Fisher Scientific	Cat# Z1376
Critical commercial assays		
Chromium Next GEM Single Cell 3' Kit v3.1	10x Genomics	Cat# 1000268
Deposited data		
Mathys et al. ²⁹	AMP-AD Knowledge Portal	Synapse:syn18485175
Zhou et al. ³⁰	AMP-AD Knowledge Portal	Synapse:syn21125841
Lau et al. ³¹	Gene Expression Omnibus (GEO)	GEO: GSE157827
Integrated human snRNA-seq dataset	This paper	Mendeley Data: https://doi.org/10.17632/f5kx9k264n.1
Mouse combination and vehicle treatment snRNA-seq dataset	This paper	Mendeley Data: https://doi.org/10.17632/f5kx9k264n.1
Experimental models: Organisms/strains		
Mouse: 5xFAD (6SJL) Tg	The Jackson Laboratory, ME, USA	MMRRC Strain# 034848-JAX
Mouse: Tau P301S (PS19)	The Jackson Laboratory, ME, USA	Strain# 008169
Other		
Chromium Controller & Next GEM Accessory Kit	10x Genomics	Cat# 1000204
Illumina NovaSeq 6000	Illumina	https://www.illumina.com/systems/sequencing-platforms/novaseq.html
Software and algorithms		
Seurat v4.0.4 (human AD dataset integration and analysis)	Stuart et al. ⁴⁰ and Hao et al. ⁸³	https://cran.r-project.org/web/packages/Seurat/index.html
Seurat v4.3.1 (mouse treatment dataset analysis)	Stuart et al. ⁴⁰ and Hao et al. ⁸³	https://cran.r-project.org/web/packages/Seurat/index.html

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
R v.4.0.3	R Core Team, 2019	http://www.R-project.org/
RStudio v.1.4.1103	R Core Team, 2019	https://cran.rstudio.com/
CellRanger v7.0.0	10x Genomics	https://github.com/10XGenomics/cellranger
dplyr: 1.1.4	R Core Team, 2019	https://cran.r-project.org/package=dplyr
ggplot2: 3.5.1	R Core Team, 2019	https://cran.r-project.org/package=ggplot2
patchwork: 1.3.0	R Core Team, 2019	https://cran.r-project.org/package=patchwork
Scran: 1.1.3	R Core Team, 2019	https://cran.r-project.org/package=scran
Scran: 1.1.3	R Core Team, 2019	https://cran.r-project.org/package=cowplot
BRETIGEA: 1.0.3	R Core Team, 2019	https://cran.r-project.org/package=BRETIGEA
Knitr: 1.49	R Core Team, 2019	https://cran.r-project.org/package=knitr
Stringr: 1.5.1	R Core Team, 2019	https://cran.r-project.org/package=stringr
dittoSeq	Bioconductor	https://www.bioconductor.org/packages/release/bioc/html/dittoSeq.html
BioRender	BioRender	www.biorender.com
g:Profiler	Reimand et al. ⁸⁴	https://biit.cs.ut.ee/gprofiler/gost
EnrichR-KG	Kuleshov et al. ⁸⁵	https://maayanlab.cloud/Enrichr/
Cytoscape	Reimand et al. ⁸⁴	https://cytoscape.org/index.html
Fiji v2.9.0	ImageJ	https://imagej.net/software/fiji/downloads
BZ-X800 analysis software – image stitching	Keyence	https://www.keyence.com/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human single nuclei RNA-sequencing (snRNA-seq) data curation

To ensure a diverse representation of AD patients, we curated publicly available human snRNA-seq datasets from three independent sources. Both Mathys et al. and Zhou et al. studies were obtained from the Accelerating Medicines Partnership Alzheimer’s Disease Project (AMP-AD) Knowledge Portal (<https://adknowledgeportal.org>) under the Religious Order Study and Memory and Aging Project (ROSMAP). The Mathys et al.²⁹ dataset is accessible through Synapse:syn18485175. The Zhou et al.³⁰ dataset is available under the study snRNAseqAD_TREM2 and is also accessible through Synapse: syn21125841. Only individuals without TREM2 mutations were included in our integrated dataset. The third dataset by Lau et al.³¹ was obtained from Gene Expression Omnibus (GEO) under the accession number GSE157827.

Case-control standardization across datasets

The Mathys et al. dataset consisted of 24 AD and 24 controls (70,634 cells and 17926 features). The Zhou et al. dataset consisted of 11 AD and 11 controls (72,489 cells and 33694 features), excluding samples with TREM2 mutations. The Lau et al. dataset included 12 AD and 9 controls (165,959 cells and 33538 features). To standardize AD identification across studies, samples were re-classified into AD or control groups based on tau tangles severity with Braak clinical staging³⁹ scores and A β burden using Consortium to Establish a Registry for Alzheimer’s Disease (CERAD) scores³⁸ as proxy. Braak staging ranges from I to VI to represent low to high levels of tau deposition. The CERAD scoring ranges from 1 to 4 to indicate high to low A β burden severity. Based on the available metadata from the original studies, we defined AD cases as individuals with severe tau deposition (Braak $\geq$ IV) and high A β load (CERAD $\leq$ 2), and non-AD controls as individuals with low tau (Braak $\leq$ III) and low A β load (CERAD $\geq$ 3). Individuals with scores that did not fulfill both criteria were excluded. Our final integrated dataset consisted of 37 AD (19 females and 18 males) cases and 38 controls (17 females and 21 males) (Figure 1A).

Cohort selections in real-world human electronic medical records (EMR)

Data in this study was provided by UC-wide OMOP EMR. <https://ctsi.ucla.edu/uc-health-data-warehouse-uchdw>.

While our drug screening approach generated a total of 86 drug candidates targeting six major brain cell types, we prioritized 25 drugs that significantly reversed AD transcriptomic signature in more than one brain cell type. This allows maximizing coverage of multiple disease-relevant cell types by combining just two drugs. The beneficial effects of the top drug hits were examined using relative risk scores established by analyzing AD prevalence (% AD diagnosis) in drug-exposed individuals and compared to propensity-matched controls, via UC-wide EMR. AD diagnosis was defined with ICD10 codes G30.0, G30.1, G30.8, and G30.9. At the time of surveying, the UC-wide EMR aggregated clinical data covering 1,441,778 individuals across six University of California (UC) campuses, encompassing more than 10 million patient records including diagnosis and medication prescriptions. For each screened

drug candidate, patients who were prescribed or administered the drug were identified using the drug exposure table and string-matching for drug names. Only individuals above the age of 65 were included in the study, as AD prevalence increases more rapidly with age after 65 years old.⁷²

The matched controls were identified using a propensity score matching approach⁵² based on these covariates: age or age at death (if applicable), race, ethnicity, sex, original indications for the drugs (for example, breast cancer was used for letrozole and colorectal cancer was used for irinotecan, also listed in Table S3 under “drug indications”), AD comorbidity (such as hypertension), and UC center locations (including UC San Francisco, UC Davis, UC Los Angeles, UC Irvine, UC San Diego, and UC Riverside). We measured the relative risk score for each drug by comparing percent prevalence of AD diagnosis (not including patients with only a diagnosis of Mild Cognitive Impairment) in the drug-exposed group and the matched control groups.

Mouse cohort generation

The 5xFAD³³ (6SJL) Tg mice (The Jackson Laboratory, ME, USA), overexpress both mutant human amyloid beta (A4) precursor protein 695 (APP) with the Swedish (KM670/671NL), Florida (I716V), and London (V717I) Familial Alzheimer’s Disease (FAD) mutation and human PS1 harboring two FAD mutations, M146L and L286V. These mice were crossed with the widely used tauopathy model, PS19³⁴ transgenic mice expressing Tau-P301S mutation and including four microtubule-binding domains and one N-terminal insert. A large cohort of the cross, including both sexes, was genotyped, and the littermates carrying transgenes from both lines were subsequently used for the dosing experiment.

METHOD DETAILS

Human snRNA-seq dataset integration, normalization, and batch correction

Before merging the datasets, cells deemed as poor quality were removed with the following criteria: total feature count less than 500, total features less than 250, or mitochondrial gene ratio higher than 10%. Sparse features, expressed in fewer than 10 cells, were also removed.

Simply merging datasets based on principle component analysis (PCA) reveals apparent study-associated variations. To perform batch harmonization while maximizing the conservation of disease-relevant biological variance, we performed canonical correlation analysis (CCA)⁴⁰ using the Seurat package v4.0.4³³ to harmonize the merged dataset. We applied the default settings to first log-normalize the merged dataset with Seurat’s *NormalizeData* function. Then, for each dataset independently, the top 2000 variable features were identified using the *FindVariableFeatures* function with “vst” as the selection method. Integration anchors were identified with the *FindIntegrationAnchors* function based on the top variable features. Finally, a harmonized dataset consisting of 179,690 cells and 29,120 features was generated using the *IntegrateData* function in Seurat.

The data matrix was linearly transformed using Seurat’s *ScaleData* function. Dimensionality reduction through Uniform Manifold Approximation and Projection (UMAP) was performed with the *RunUMAP* function, which considers the top 30 dimensions selected from the corresponding principal component analysis (PCA) obtained by running the *RunPCA* function in Seurat.

Clustering was determined based on the first 30 PCs using the *FindNeighbors* function in Seurat, which embeds cells in a K-nearest neighbor graph based on Euclidean distance in PCA space and refines the edge weights between any two cells based on the shared overlap in their local neighborhoods. Clustering was implemented using a resolution of 0.5 in the *FindClusters* function, which applies modularity optimization techniques such as the Louvain algorithm, resulting in a set of 27 distinct clusters.

Human cell type annotation

Cell type identities were determined by applying Seurat’s *AddModuleScore* function to lists of known human brain marker genes (~8 per cell type) collected from PanglaoDB⁸⁶ and referenced by Jiang et al.⁸⁷ Cell type assignment included astrocytes, microglia, oligodendrocytes, oligodendrocyte precursor cells, endothelial cells, pericytes, and excitatory and inhibitory neurons. Each cell was assigned the corresponding cell type identity that generated the highest scores among scores for all cell types. If the highest and second highest scores of a cell were within 20% of the highest score, then the cells were deemed hybrids and excluded from further analysis. Then, for each cluster, cell type identity was manually annotated based on the dominant assigned cell type in each cluster; highly heterogeneous clusters were labeled unresolved and excluded, yielding 21 distinct clusters in the integrated dataset. We assessed the validity of the assigned cell type identities by examining the homogeneity, distribution, and separation of cell types clustering in UMAP plots.

QUANTIFICATION AND STATISTICAL ANALYSIS

Cell-cell communication analysis

Cell-cell communication (CCC) was calculated using the *CellChat* method.⁸⁸ Briefly, *CellChat* utilizes ligand, receptor, and cofactor expression from transcriptomic data to calculate a CCC probability. First, based on a *CellChat*-curated database of ligand-receptor interactions, differentially expressed signaling genes are used to calculate the ensemble average expression of signaling genes. Communication probability is modeled using the law of mass action, and statistically significant communications are identified using a permutation test. We then evaluated the interaction across cell types (excitatory neurons, inhibitory neurons, oligodendrocytes,

oligodendrocyte precursor cells, microglia, and astrocytes). For the mouse vehicle-treatment comparison, and the human control-AD comparison, the *netVisual_diffInteraction* was used to calculate the net differential interaction strength between each of the two groups.

Cell-type-specific differential expression analysis

For each cell type, we performed differential gene expression analysis comparing AD samples to controls using the *FindMarkers* function in the Seurat package. We set the `test.use` parameter to MAST,⁸⁹ which uses a two-part generalized linear model that models gene expression rate using linear regression and expression level using Gaussian distribution, as recommended by Mou et al.⁹⁰ in a study comparing 9 differential expression analysis methods for single-cell RNA sequencing datasets. We determined differentially expressed genes (DEGs) as those with adjusted p-value < 0.05 based on Bonferroni correction using all genes in the dataset and a log2 fold change (logFC) greater than 0.1.

Pathway enrichment analysis

Pathway enrichment analysis was conducted using g:Profiler, a web tool that performs functional over-representation analysis by mapping a given gene list to known functional information sources and detects statistically significant enriched terms.⁸⁴ As input gene lists, we queried each cell-type-specific DEG list independently with an adjusted p-value cutoff of 0.05 to obtain significant pathway enrichment in both directions. Multiple testing correction was performed using the g:SCS algorithm, which specifically corrects for p-values obtained from GO and pathway enrichment analysis.⁸⁴ This method is designed to correct for multiple tests that may be potentially dependent on each other due to term associations. In addition to considering Gene Ontology (GO) cellular components, biological processes, and molecular functions as supplementary data, we focused our analysis on the Kyoto Encyclopedia of Genes and Genomes (KEGG) functional pathways.

Computational drug repurposing analysis

The computational drug repurposing algorithm, which was developed by Sirota et al.,⁹¹ and taken from Chen et al.,⁴⁸ was applied to each disease gene signature profile using the publicly available Connectivity Map (CMap) database, consisting of treatment profiles of more than 1300 FDA-approved drugs or previously investigated compounds. The drug repurposing algorithm we employed takes two inputs: 1) an ordered list of up and down-regulated genes from individuals with disease as compared to controls and 2) the data from CMap, consisting of rank FC of each gene after drug treatment relative to vehicle controls on the same plate. The pipeline assesses the disease-drug relationship using CMap scores derived from a Kolmogorov-Smirnov test (K-S test), comparing gene expression ranks in disease and by a drug. The pipeline was adapted for single-cell analysis, where each cell-type-specific DEG signature set was overlaid with 6100 drug profiles on the 1300 drugs provided by the CMap. A drug with a strong negative CMap score indicates an opposing mechanistic relationship with the disease, suggesting therapeutic potential by reversing the regulation direction of disease signatures. The absolute values of CMap scores reflect the degrees to which the drug “flips” the signature of the disease. To address variations in input counts across cell types, significantly reversed drug profiles were identified for each cell type separately using a permutation-based approach. The false discovery rate (FDR; Benjamini-Hochberg) was calculated to adjust p-values. P-values for individual drug hits were determined by comparing reversal scores to a distribution of random scores that were generated by the permutation strategy. Negative reversal scores were deemed significant if they met the criterion of FDR < 0.05. For drugs tested multiple times (e.g., in different cell lines), the profile with the most substantial reversal (lowest negative score) was used.

AD outcome measures for drug candidates using UC-wide EMR

We employed the R package, MatchIt,⁵³ and used a nearest-neighbor matching algorithm with a 1:2 matching scheme, where each drug-exposed individual was propensity-score matched with two unexposed controls sharing similar covariate characteristics. To assess covariate balance, standardized mean differences between the drug-exposed and control groups before and after propensity score matching were calculated (Figure S2). Bootstrapped χ -squared tests across 10 permutations of the iterative matching of the control group were applied to establish significance. Drugs with a significantly reduced risk of AD were included in Figure 2E. Other drugs found in the UC-wide EMR but did not yield significance or reduced risk were reported in Table S3. Individualized original indications were used in matching for each drug.

Drug selection rationale for validation in AD mouse model

Our integrated human single-cell transcriptomic analysis revealed two distinct pathological clusters among the six cell types analyzed. Additionally, we observed that AD inhibitory neurons exhibited the most pronounced disparity between AD cases and controls, underscoring its significance in AD pathology. These discoveries informed our prioritization of drug candidates targeting these specific clusters and accommodating inhibitory neurons for treating AD. Among a selection of five promising drug candidates that demonstrated a significant reduction in AD risk in humans, we prioritized letrozole and irinotecan, since letrozole targets the neuronal pathological cluster while irinotecan targets the glial pathological cluster. We hypothesize that the combination of letrozole and irinotecan holds synergistic therapeutic effects, with each pathological cell type cluster addressed by one EMR-validated drug candidate. Consequently, we examined AD outcomes in patients receiving both drugs under their original disease indications

in the UC-wide EMR. Unfortunately, fewer than 100 patients were ever prescribed both medications, rendering statistical analysis inconclusive. To explore the synergistic effects of this combination in the context of AD, we turned to direct experimentations in animal models.

Drug treatments

The transcriptomic signatures in CMap were profiled with a short time treatment of 6 hours in the prostate cancer cell line PC3 at 14 μ M for letrozole and in the breast cancer epithelial cell line MCF7 at 100 μ M for irinotecan.³² These concentrations are much higher than what is used in humans. To mimic human dosage from EMR and to ensure tolerable doses for long-term treatment regimens, exceeding 3 months with every-other-day administration (to minimize stress by reducing injection frequency), we selected concentrations of 1 mg/kg for letrozole and 10 mg/kg for irinotecan. These doses are equivalent to human dosages and within ranges used in prior mouse studies for cancer, while maintaining an approximately 10:1 ratio of irinotecan to letrozole used in CMap.

According to Arora et al.,⁵⁷ letrozole concentration reaches 1 μ g/mL in the mouse brain after an 8mg/kg intraperitoneal (IP) injection. For letrozole (molecular weight: 285.31 g/mol), to achieve a 14 μ M brain concentration from CMap (14 μ M letrozole = $(14 \times 10^{-6} \text{ mol/L}) \times (285.31 \text{ g/mol}) = 0.003994 \text{ g/L} = 3.994 \text{ }\mu\text{g/mL}$), approximately 32 mg/kg needs to be given. This is significantly higher than the dosing range (0.2 - 2mg/kg) in cancer studies using mouse models.⁵⁸ Letrozole treatment in humans is given daily at a dose of 2.5 mg⁵⁰ (based on an average weight of 65kg, the daily dose in humans is approximately 38 μ g/kg. To determine the mouse equivalent dose, we used animal dose(mg/kg) = human dose (mg/kg) * (human Km factor / mouse Km factor), and according to the FDA guidelines, the Km factor for mice is 3, and for humans is 37.⁹² So, mouse dose = $0.038 \times (37/3) = 0.47 \text{ mg/kg}$ daily. Therefore, for an every-other-day injection regimen, we gave letrozole at $0.47 \text{ mg/kg} \times 2 = 0.94 \text{ mg/kg} \sim 1 \text{ mg/kg}$.

For irinotecan (molecular weight of irinotecan hydrochloride (CPT-11) = 623.14 g/mol), a 1 μ g/mL concentration in mouse brain was reported with 40mg/kg IP injections from a study by Goldwirth et al.⁵⁹ To achieve 100 μ M brain concentration from CMap ($100 \times 10^{-6} \times 623.14 = 0.062314 \text{ g/L} = 62.314 \text{ }\mu\text{g/mL}$), approximately 2493 mg/kg needs to be given, which is also significantly higher than the dosing range (4 - 160 mg/kg) in cancer studies using mouse models. Irinotecan is given to humans at 125 mg/m² weekly,⁹³ equivalent to 3.38mg/kg, assuming $1 \text{ m}^2 \approx 37 \text{ kg}$ body weight, and about 483 μ g/kg daily. Mouse dose = $0.483 \times (37/3) = 5.96 \text{ mg/kg}$ daily dose, and irinotecan was given at 10 mg/kg every other day, approximately double the daily dose.

A solution of either letrozole (1 mg/kg) or irinotecan (10 mg/kg) alone, or of the two combined was prepared weekly in 0.9% sterile saline containing 10% Tween-20 to obtain the final concentrations. All treatment solutions were sonicated for 30 minutes before injections to ensure proper mixing of the drugs. The treatments were given by i.p. injection every-other-day to the 5xFAD/PS19 double transgenic cohort, evenly split into four groups and sex-balanced, starting 12 weeks before and continuing throughout the behavioral assessment. Mice were aged 4-5 months old before the start of the treatment and the treatments lasted about 4 months until sacrificed. Body weight was measured weekly during drug treatment to ensure no significant drop in body weights due to drug toxicity; injection volume was calculated based on body weight. Control mice were injected with a matched volume of vehicle, 10% Tween-20 in 0.9% sterile saline at pH 8.5. Injections were well tolerated and had no adverse effects on health.

Behavioral test and analysis

Mice were housed with littermate controls. Each mouse was assigned a random number, so researchers were blinded to genotype and treatment information. Male and female mice were tested in separate rooms with the same settings and test duration to avoid olfactory cues becoming a distraction with testing a mixed sex cohort on the same equipment.

The Morris Water Maze pool (diameter, 122 cm) contained opaque water ($21 \pm 13^\circ\text{C}$) with a platform 10 cm in diameter. Mice were first trained for 6 days to locate a hidden platform submerged 1.5cm below the surface of the water using distal cues surrounding the pool. These 6 days consisted of two training sessions, 2 hours apart, each consisting of two trials with a maximum latency of 60s, with a 15-minute inter-trial interval. Entry points were changed semi-randomly between trials, but distal visual cues on the walls of the behavioral testing room remained constant throughout the test. To test for memory retention, at 24 and 72 hours after the last hidden platform training, a 60-s probe trial (platform removed) was done. The entry point for the probe trial was opposite to the target. Finally, as a control for visual acuity and motor ability, mice were tested with a cued platform where the submerged platform is indicated by a black and white striped mast 15cm high extending out the surface of the water. The platform location remained constant in the hidden platform sessions but was changed for each visible platform session. Three sessions of two trials, with a maximum latency of 60s, was performed over 2 days. Performance was monitored with an EthoVision video-tracking system (Noldus Information Technology). For the probe trials, we analyzed (1) target quadrant preference- the percent time spent in the target quadrant versus average time spent in the three other quadrants, (2) platform crossing- the number of crossings over the position of the target platform versus the average number of crossings over the equivalent positions in the three other quadrants as well as (3) escape latency- a memory probe measuring how fast mice arrive at the platform location after placement in the maze. An ANOVA was used to analyze the effect of treatment, genotype, and probe timing on percent on-target crosses.

Histopathological analyses

One hemibrain per mouse was drop-fixed in 4% paraformaldehyde prepared in 1xPBS, washed for 24 hours in 1xPBS, then cryo-protected in 30% sucrose for 48 hours at 4°C . The fixed hemispheres were cut into 30 μ m thick coronal slices using a freeze-sliding

microtome (Leica). These sections were then stored in a cryoprotectant solution of 30% ethylene glycol, 30% glycerol, and 40% 1xPBS at -20°C .

Hippocampal brain sections (ten sections per mouse spaced approximately 300 μm apart, 30- μm thick) were mounted onto microscope slides from Fisher Scientific. A 0.1% Sudan Black solution was prepared by dissolving Sudan Black powder (Sigma) in 70% ethanol (KOPTEC) and mixing it with a magnetic stirrer. After centrifuging the solution at 1,100g for 10 minutes, the supernatant was filtered using a 0.2- μm filter syringe (Thermo Scientific). The brain sections were stained with the 0.1% Sudan Black solution for 10 minutes, then washed in 70% ethanol followed by Milli-Q water. The sections were then coverslipped with ProLong Gold mounting medium (Invitrogen). All sections were imaged; eight consecutive brain sections with consistent anatomic locations from AP -1.2 to -3.6 were quantified.

For Thio-S staining, several brain sections spaced 300 μm apart were mounted onto slides, following a protocol adapted from a previous study. The sections were washed with $1\times$ PBS-T and then incubated in a solution of 0.06% Thio-S in PBS for 8 minutes. After this incubation, the sections were washed for 1 minute in 80% ethanol and then for 5 minutes in PBS-T. The sections were counterstained with DAPI for 8 minutes, washed again with PBS-T, and coverslipped. Three sections per mouse were quantified and averaged for percent Thio-S positive area and number of plaque counts (normalized by hippocampal area size per section).

For immunofluorescent staining, several brain sections spaced approximately 300 μm apart, were washed three times with $1\times$ PBS-T (PBS + 0.1% Tween-20) (Millipore Sigma) and then incubated for 5 minutes in boiling antigen retrieval buffer (Tris buffer, pH 7.6; TEKNOVA). After this, the sections were rinsed in PBS-T before being placed in a blocking solution composed of 5% normal donkey serum (Jackson Labs) and 0.2% Triton-X (Millipore Sigma) in $1\times$ PBS for 1 hour at room temperature. Next, the sections were washed again in PBS-T and incubated in Mouse-on-Mouse (MOM) Blocking Buffer (one drop MOM IgG in 4 ml PBS-T) (Vector Labs) for another hour at room temperature. Following the MOM block, the sections were incubated overnight at 4°C in primary antibodies diluted to their optimal concentrations. The antibodies and their dilutions included: anti-AT8 (ms, 1:300, Invitrogen), anti-GFAP (ms, 1:500, Millipore Sigma), anti-Iba1 (rbt, 1:300, Wako), anti-NeuN (GP, 1:500, Millipore Sigma). Three sections per mouse were quantified and averaged for a positive percent area.

After the primary antibody incubation, the sections were washed in PBS-T and then incubated for 1 hour at room temperature in secondary antibodies. These included donkey anti-mouse 488 (1:1,000, Abcam), donkey anti-rabbit 594 (1:1,000, Abcam), and donkey anti-guinea pig 594 (1:1,000, Jackson Immuno). Subsequently, the sections were washed in PBS-T and incubated in DAPI (1:30,000 dilution in PBS-T) (Thermo Fisher) for 8 minutes at room temperature. After a final wash with PBS-T, the sections were mounted onto microscope slides (Fisher Scientific), coverslipped with ProLong Gold mounting medium (Vector Laboratories) and sealed with clear nail polish.

Images for quantifications were captured using a scanning microscope (Keyence) at magnifications of $\times 10$ or $\times 20$, depending on the stain. To minimize batch-to-batch variation, all samples for each stain were processed simultaneously and imaged at the same fluorescent intensity. For quantifying the percent coverage area, an optimal threshold was established for each stain in Fiji, and all samples were quantified using this threshold. To prevent bias, researchers were blinded to the sample identities. Representative images were captured using an Aperio VERSA slide scanning microscope (Leica) at $\times 20$ magnification.

snRNA-seq library preparation and sequencing

The other hemibrains were dissected by brain subregions, rapidly frozen on dry ice, and kept at -80°C . Hippocampal samples were used for single-nuclei preparation. One frozen mouse hippocampus was placed into a pre-chilled 2 mL Dounce with 1 mL of cold 1X Homogenization Buffer (1X HB) (250 mM Sucrose, 25 mM KCL, 5 mM MgCl_2 , 20 mM Tricine-KOH pH7.8, 1 mM DTT, 0.5 mM Ser-midine, 0.15 mM Serpine, 0.3% NP40, 0.2 units/ μL RNase inhibitor, ~ 0.07 tabs/ml cOmplete Protease inhibitor). Dounce with “A” loose pestle (~ 10 strokes) and then with “B” tight pestle (~ 10 strokes). The homogenate was filtered using a 70 μm Flowmi strainer (Bel-Art) and transferred to a pre-chilled 2 mL LoBind tube (Fisher Scientific). Nuclei were pelleted by spinning for 5 min at 4°C at 350 RCF. The supernatant was removed, and the nuclei were resuspended in 400 μL 1X HB. Next, 400 μL of 50% Iodixanol solution was added to the nuclei and then slowly layered with 600 μL of 30% Iodixanol solution under the 25% mixture, then layered with 600 μL of 40% Iodixanol solution under the 30% mixture. The nuclei were then spun for 20 min at 4°C at 3,000RCF in a pre-chilled swinging bucket centrifuge. 200 μL of the nuclei band at the 30%-40% interface was collected and transferred to a fresh tube. Then, 800 μL of 2.5% BSA in PBS plus 0.2 units/ μL of RNase inhibitor was added to the nuclei and then were spun for 10 min at 500 RCF at 4°C . The nuclei were resuspended with 2% BSA in PBS plus 0.2 units/ μL RNase inhibitor to reach at least 500 nuclei/ μL . The nuclei were then filtered with a 40 μm Flowmi cell strainer. The nuclei were counted and then $\sim 13,000$ nuclei per sample were loaded onto 10x Genomics Next GEM chip M. The snRNA-seq libraries were prepared using the Chromium Next GEM Single Cell 3' HT kit v3.1 (10x Genomics) according to the manufacturer's instructions. Libraries were sequenced on an Illumina NovaSeq 6000 sequencer at the UCSF CAT sequencing core.

Sequence alignment, filtering, and counting

The demultiplexed fastq files were aligned to the mouse reference genome, mm10-1.2.0, which includes introns, using the *cellranger count* function (version 4.0.0) with default parameters, as detailed in the Cell Ranger documentation. Subsequently, a single UMI count file per animal/sample was generated by the *cellranger count* function. Individual UMI count files were then combined into

a single count matrix using the *merge* function in the Seurat package v4.3.01. Metadata, including age, sex, and treatment information, were added to each cell.

Pre-processing and quality control

The count matrix was further processed with Seurat by first calculating the percentage of mitochondria genes mapped per cell. The distribution of feature count, total mapped gene count, and percentage of mitochondria genes were visualized across biological samples as violin plots, and no obvious outlier was identified. We filtered the count matrix to only include cells with higher than 250 gene features, at least 500 gene counts, and mitochondria gene percentages lower than 1%. Potential misaligned or ambiguous gene features expressed in fewer than 10 cells were also removed. These quality assurance steps resulted in a final Seurat object containing 25,642 gene features expressed by 237,853 nuclei.

Normalization, dimensional reduction, and clustering

Count normalization and dimensionality reduction were conducted following standard procedure in the Seurat package v4.3.01. In brief, we performed normalization and variance stabilization with an updated version of *sctransform*, v2, and principal component analysis (PCA) with *RunPCA* (npcs = 30). Dimensional reduction through Uniform Manifold Approximation and Projection (UMAP) was performed with the *RunUMAP* function and considering the top 10 dimensions selected from the corresponding PCA.

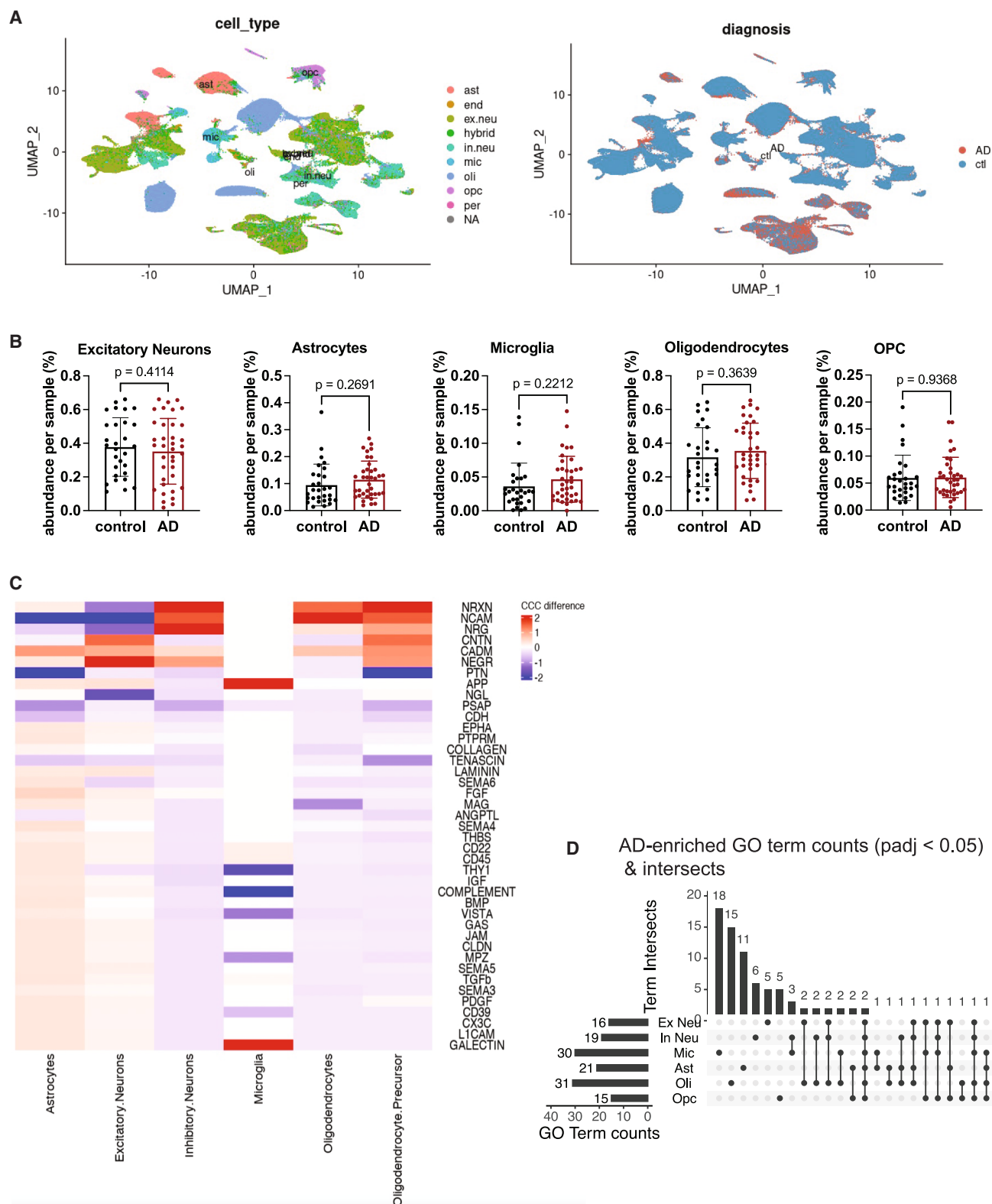
The clustering was based on the first 10 principal components (PCs) and resolution of 0.5 using the *FindNeighbors* function in Seurat. This function embeds cells in a K-nearest neighbor graph, considering the Euclidean distance in PCA space and refining the edge weights between any two cells based on the shared overlap in their local neighborhoods. The clustering was obtained using the *FindClusters* function, which employs modularity optimization techniques such as the Louvain algorithm, with a resolution parameter of, resulting in a set of 31 distinct clusters.

Mouse cell type annotation, differential gene expression, and pathway enrichment analysis

Major cell types, including astrocytes, microglia, oligodendrocytes, oligodendrocyte precursor cells, and excitatory and inhibitory neurons, were classified using mouse brain cell markers in PanglaoDB,⁸⁶ a publicly available marker gene database. Further subdivisions of hippocampus cell types, such as CA1 and CA3 pyramidal cells, were queried against hippocampal cell-type-specific marker genes as published in hipposeq (<https://hipposeq.janelia.org>). Cell type identities per cluster were determined by applying Seurat's *AddModuleScore* function to sets of mouse brain marker genes. A module score for each cell type considered was calculated per cell. Each cell was assigned the corresponding cell type identity that generated the highest scores among scores for all cell types. If the highest and second highest scores of a cell were within 10% of the highest score, then the cells were deemed hybrids and excluded from further analysis. We assessed the validity of the assigned cell type identities by examining the homogeneity, distribution, and separation of cell types by clustering in UMAP plots. Each cluster is annotated with the dominated cell type identity. Minority cell types in a cluster, defined as a cell type that accounts for less than 5% of the total counts for that cluster, were considered potential hybrid cells and excluded from further analysis.

For the cell-type-specific differential gene expression analysis, we used the same procedure and tools in the human-integrated analysis. For pathway enrichment analysis, in addition to using g:Profiler, we curated the EnrichR-KG web tool,⁸⁵ which facilitates analysis across multiple databases and provides visual representations linking significantly enriched genes with the associated pathways and GO terms. The reversal gene-pathway network was generated and downloaded using EnrichR-KG with significantly differentially reversed genes as inputs. For each cell-type-stratified analysis, the top five KEGG pathways and GO terms were displayed in a network format.

Supplemental figures



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Figure S1. Additional UMAP, cell type abundance, cell-cell-communication, and enriched pathway results in the integrated human single-nucleus transcriptomic dataset, related to [Figure 1](#)

- (A) UMAP plots of merged samples before CCA integration, labeled with cell type identities and disease status.
- (B) Scattered bar plots of cell type abundance (percentage) for the other five cell types, comparing AD and controls (unpaired two-sided t test was used. Data are represented as mean $\pm$ SEM).
- (C) Heatmap displaying pathway-level cell-cell-communication (CCC) difference comparing AD and controls and stratified by cell type. Color value represents the scaled CCC probability difference in AD compared with controls. Signaling pathways represent ligand-receptor pairs based on the KEGG pathway database.
- (D) UpSet plot showing GO term counts for the six major brain cell types and intersects across cell types.

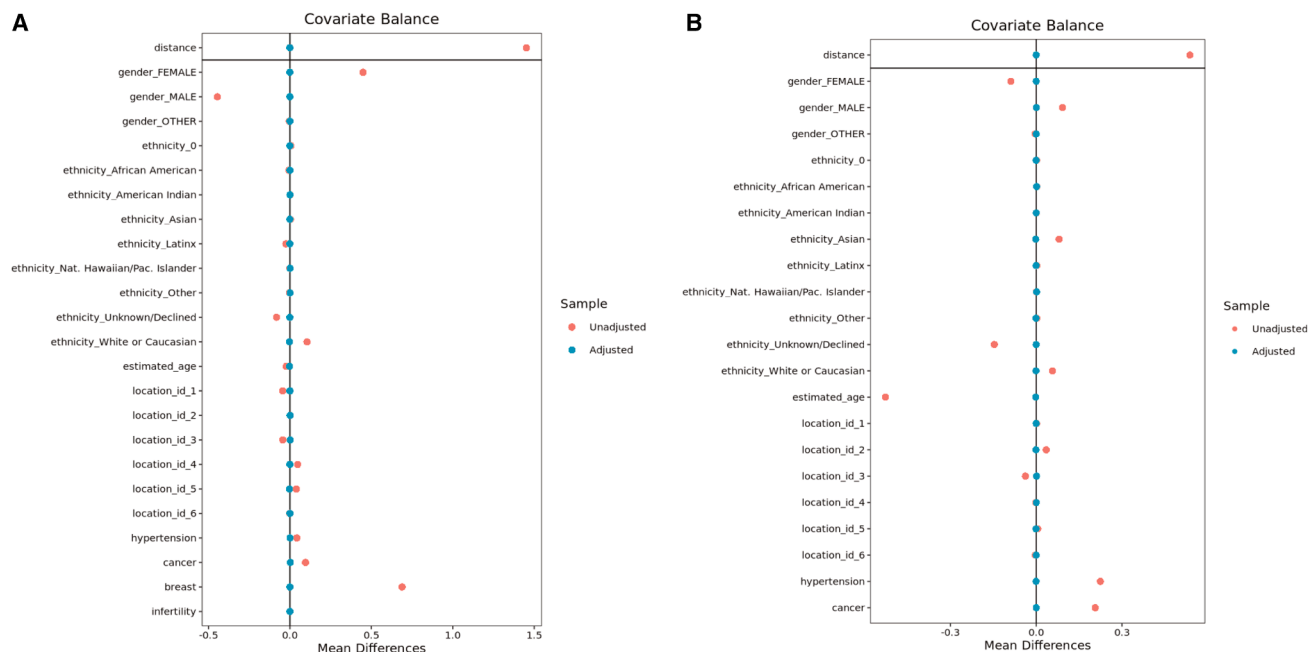


Figure S2. Matching efficiencies of letrozole and irinotecan in a population-based analysis using UC-wide EMR, related to Figure 2 (A and B) Propensity-matching results for letrozole (A) and irinotecan (B) in AD outcome measures. Covariates were balanced after adjustment using MatchIt with absolute mean difference less than 0.1 for all variables included in the matching algorithm.

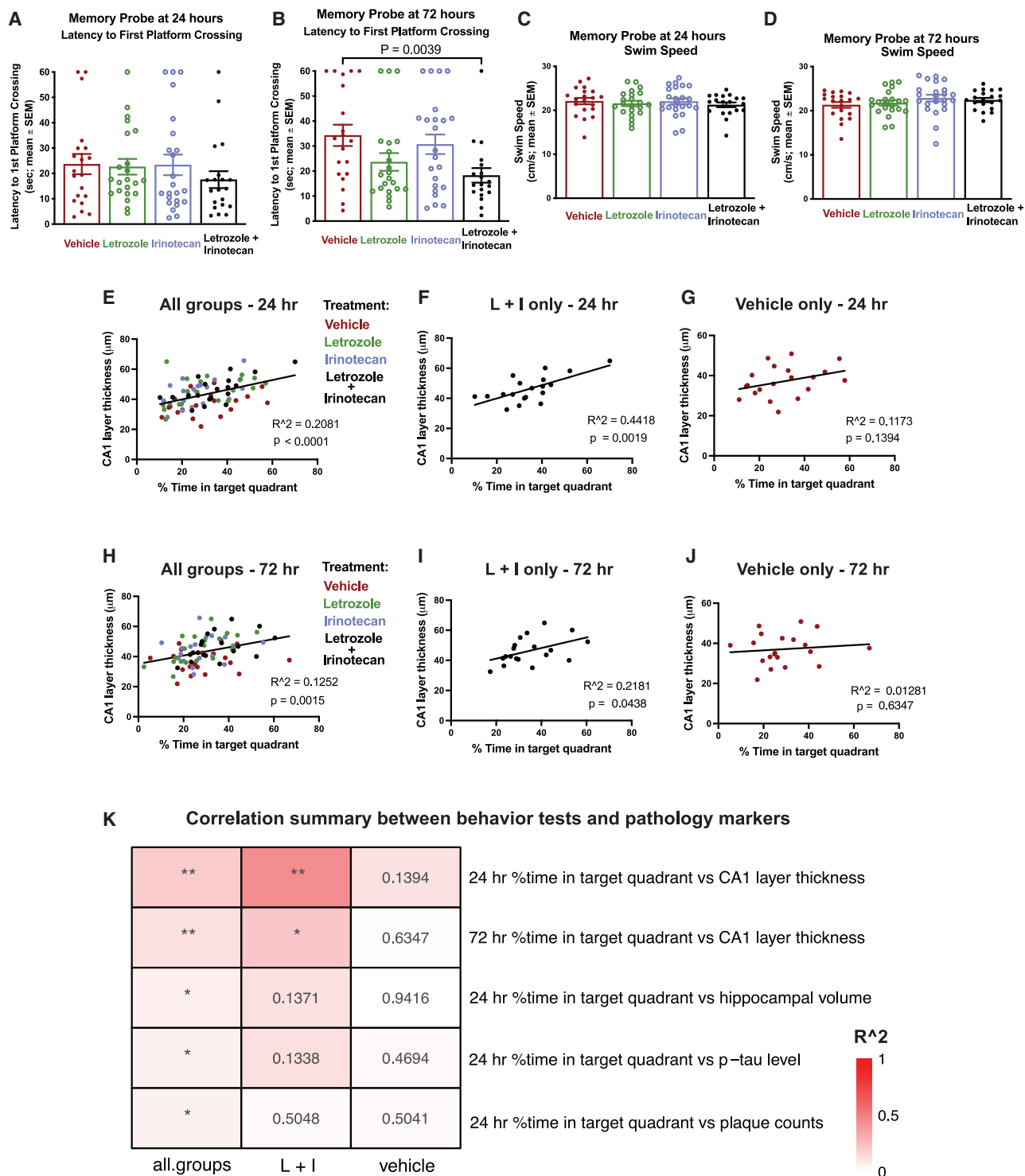


Figure S3. Additional memory probe assessments and correlation analysis between behavior and pathological assessments of treatment response in aged 5x FAD/PS19 mice, related to Figures 3 and 4

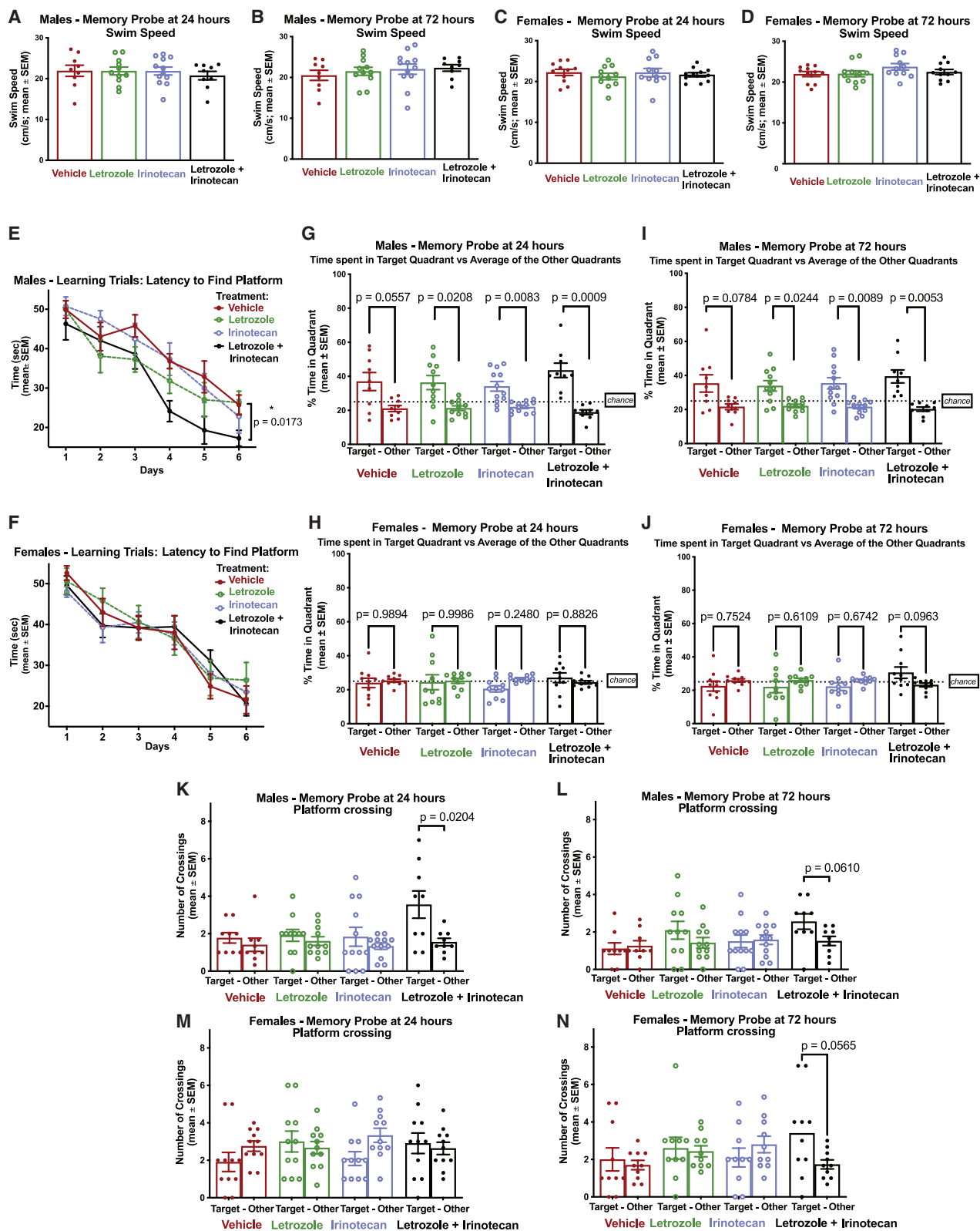
(A and B) Memory probes measuring latency to first time of platform crossing at 24 h (A) and 72 h (B) demonstrated long-term location memory rescue in combination-treated mice only, at 72 h after removing the platform.
(C and D) Swim speed at 24 h (C) and 72 h (D), no significant difference in swim speed across groups, indicating that the observed behavioral differences were not confounded by visual or motor impairment.

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(E–J) Correlations between results from memory probes and pathological measures. Time spent in target quadrant at 24 h versus NeuN-positive CA1 layer thickness of the entire experimental cohort (E), in combination-treatment group only (F) and in vehicle-treatment group only (G). Time spent in target quadrant at 72 h versus NeuN-positive CA1 layer thickness of the entire experimental cohort (H), in combination-treatment group only (I) and in vehicle-treatment group only (J).

(K) Heatmap summarizing significant correlations between behavior test results and pathology measurements in all groups, in combination-treatment group only or in vehicle-treatment group only. Color intensities resemble R square values; number and * label p values. * $p < 0.05$ and ** $p < 0.01$.

In (A)–(D), ordinary one-way ANOVA was applied between all treatment groups and vehicle control group, and p values were adjusted by Dunnett's multiple comparison test. Data are represented as mean $\pm$ SEM.



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Figure S4. Sex difference in learning and memory probe assessments of treatment response in aged 5×FAD/PS19 mice, related to Figure 3

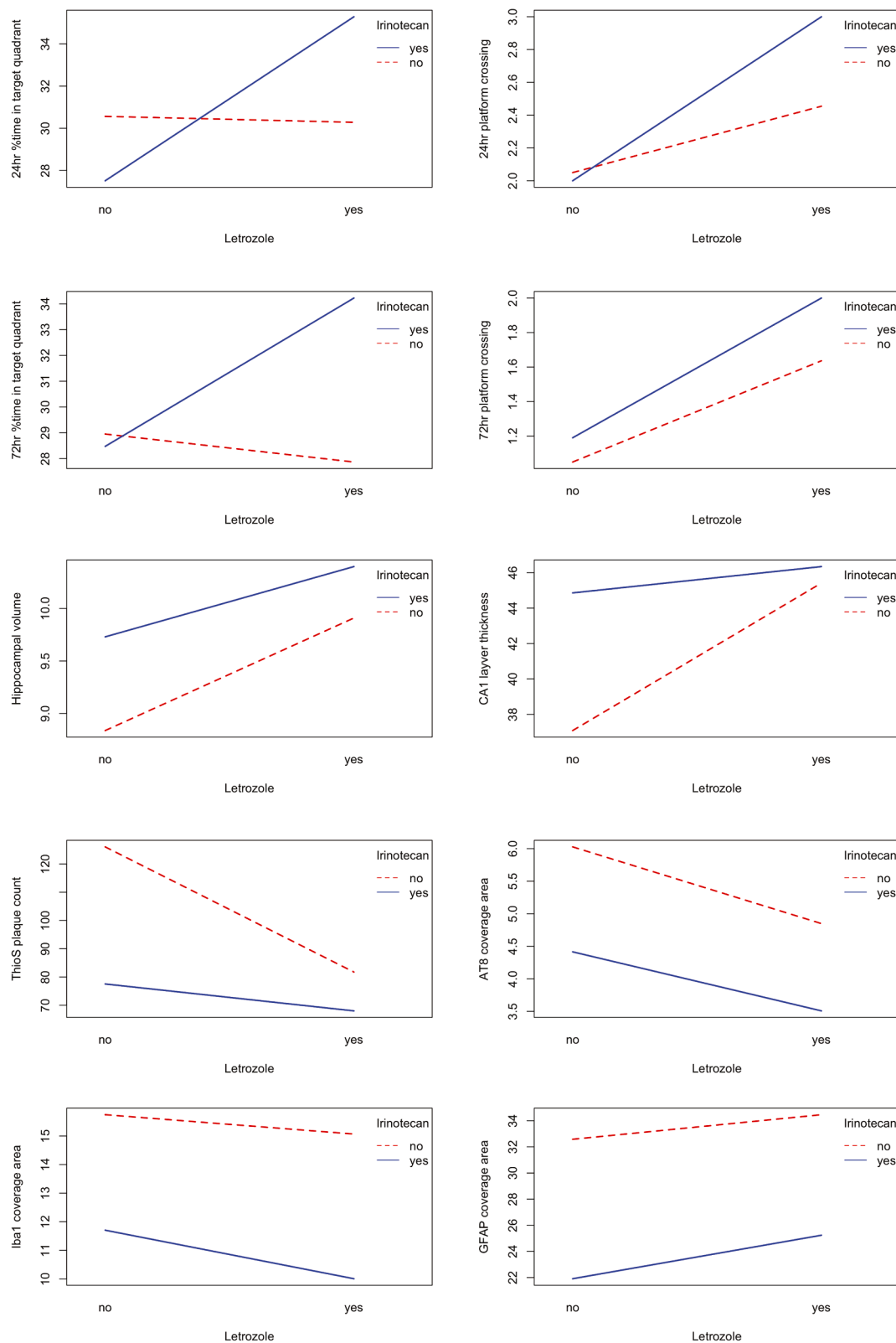
(A–D) Swim speed at 24 h (A and C) and 72 h (B and D) in male (A and B) and in female (C and D) mice. No significant difference in swim speed across groups, indicating that the observed behavioral differences were not confounded by visual or motor impairments. Ordinary one-way ANOVA was applied between all treatment groups and vehicle control group, and *p* values were adjusted by Dunnett's multiple comparison test. Data are represented as mean ± SEM.

(E and F) Escape latency over learning days 1–6 in male (E) and female (F) mice demonstrating significant better learning in combination-treated males. One-way repeated-measures ANOVA test was applied comparing vehicle group with treatment groups with Sidak's multiple comparison test. Data are represented as mean ± SEM.

(G–J) Memory probes of percent time spent in target quadrant versus average of other quadrant, demonstrating a significant preference of the target quadrant by letrozole, irinotecan, and combination-treated males but not the vehicle-treated males at 24 h (G) and 72 h (I) after removing the hidden platform. No significant preference was observed in females at 24 h (H) and 72 h (J).

(K–N) Memory probes measuring number of platform-location crossings in the target quadrant versus average of the other quadrants in male (K and L) and female (M and N) mice. Significantly more crossings in the target quadrant where the platform used to be only observed in the combination-treatment group at 24 h (K) in males, while there was a trend toward significant difference in males (L) and in females (N) at 72 h after the platform was removed.

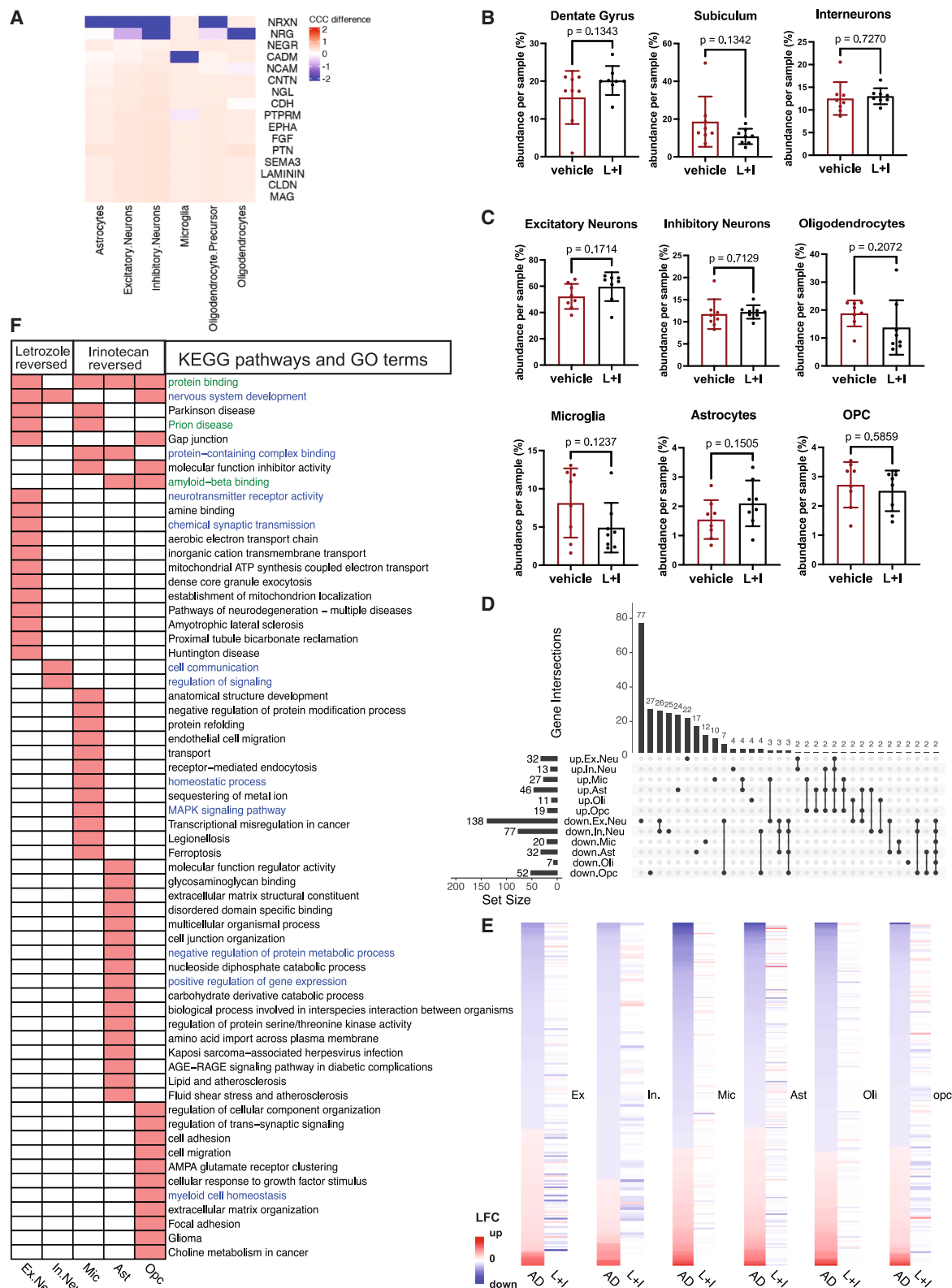
In (G)–(N), ordinary one-way ANOVA test was performed between target and other quadrants for each group, and *p* value was adjusted with Bonferroni multiple-comparisons testing. Data are represented as mean ± SEM.



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Figure S5. Letrozole and irinotecan interactive effect analysis on behavioral and pathological treatment response in aged 5×FAD/PS19 mice, related to [Figures 3 and 4](#)

Interaction plots reveal the interactive patterns between letrozole and irinotecan in their effects influencing the panel of behavioral and pathological measurements.



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Figure S6. Additional details on snRNA-seq analysis in aged 5×FAD/PS19 mice across treatment groups and pathway enrichment analysis of drug-reversed AD signatures profiled in CMap, related to Figures 5 and 6

(A) Heatmap displaying pathway-level CCC difference between combination-treated and vehicle-treated groups, and stratified by cell type. Value represents the scaled CCC probability difference in combination-treated compared with vehicle-treated groups. Signaling pathways represent ligand-receptor pairs based on the KEGG pathway database.

(B and C) Scattered bar plots of cell type abundance analysis for neuronal subtypes of hippocampus (B) and for the 6 major brain cell types (C), comparing combination-treated and vehicle-treated groups (unpaired two-sided t test; data are represented as mean ± SEM).

(D) UpSet plot showing DEG counts (combination-treated compared with vehicle-treated groups, $\text{abs}[\log\text{FC}] > 0.1$ and $\text{padj} < 0.05$) and intersects across groups (separated by cell types and direction of regulation).

(E) Heatmaps of cell-type-specific AD signature gene expression profiles (AD versus controls DEGs in the human-integrated analysis) in comparisons with gene expression changes by L + I treatment (L + I-treated versus vehicle-treated groups) in aged 5×FAD/PS19 mice across the six major cell types. Red indicates upregulation either in AD or by L + I treatment whereas blue indicates downregulation. White indicates the AD signature genes have no expression difference in L + I treatment compared with vehicle treatment or were not expressed in mice.

(F) GO terms and KEGG pathways enriched by letrozole-reversed AD signatures in excitatory and inhibitory neurons or irinotecan-reversed AD signatures in microglia, astrocytes, and OPCs based on CMap profiles. AD signature genes act in reverse manner, as drug profiles in CMap were used as input in g:Profiler to identify enriched GO driver terms and KEGG pathways. Proteopathy-related pathways were highlighted in green, whereas cell-type-specific functional pathways were highlighted in blue.